Annex VIII — Perceived Real-Time Visual Information Delivery

BEAP™ / WR Desk™ / Optirando™ Canonical Specification — standalone Annex. Version 3.1 (Final).

Document status: Normative unless explicitly marked as informative.

Applies to: WR Desk™ runtime, all Presentation Surfaces, the Composition Engine, composition libraries in all provenance classes, the Session Composition Pool, Recycled Layout Assets and the Priority Composition Signature, the Prepared Edition and all frame-structured editions, the Augmented Overlay, responsive HTML export, raster derivation, WR Anchor export and integration, and all generation profiles defined by this Annex.

Dependencies: WR Graph™ structure, non-pre-satisfaction, routing-state exclusivity, and graceful degradation (Annex II); PoAC™ governance, evidence, mutation classes, and the Cloud Abstraction Boundary (Annex III); adapter and distribution governance (Annex IV); media reconstruction and converter governance (Annex V); WR Advance Preparation, WR Predictive Preparation, Real-Time Support, the Debrief, and retrieval privacy (Annex VI); publisher artifact, manifest, grant, and Private Distribution Path machinery, including the Merkle manifest model (Annex VII); and the initiation layer, invitation classes, and public handshake (Annex IX).

Trademark notice: Optirando™, WR Desk™, WR Graph™, BEAP™, PoAE™, PoAC™, WRVault™, TimesDesk™, and WR Code® are used throughout as product and protocol designations of the Optirando™ system.

VIII.0 Status, Scope, and Relationship to the Canon

This Annex specifies Hybrid Composition: the production of visually rich information artifacts — infographics in particular — by reasoning over information and composing reusable, annotated, quality-controlled elements, rather than synthesizing every artifact as one monolithic pixel image. It further specifies:

1. the Element Library and its governed optimization lifecycle;
2. the Composition Description and Composition Engine;
3. prepared, pooled, recycled, and fallback composition paths;
4. operator-selectable generation profiles;
5. responsive HTML as the primary structured Delivery Form;
6. deterministic raster and document derivations from the same canonical composition;
7. frame-structured, priority-ranked editorial presentation and the Bayesian Priority Engine;
8. the Priority Composition Signature and Layout Recycling;

9. WR Anchor export: contract-bound integration of composed content into third-party WR-connected pages, with the complete asset set committed in the handshake Merkle manifest and hosted locally on the receiving server;
10. the versioned content update path for integrated content;
11. lifecycle-efficiency measurement from creation through consumption;
12. the Prepared Edition and Augmented Overlay.

The Annex distinguishes two layers that SHALL NOT be conflated:

- the **general composition architecture**, which may be implemented by any visual-generation or large-language-model system and may permit bounded generation of missing visual elements at request time; and
- the **WR governance profile**, in which privacy, provenance, reproducibility, and deterministic presentation impose stricter defaults while preserving informed operator choice.

This Annex does not redefine wire formats, capsule serialization, cryptographic primitives, transport, PoAE™ execution semantics, the WR Preparation lifecycle, or the media reconstruction model. It reuses those mechanisms and does not weaken them. Everything specified here is preparation, composition, delivery, or presentation. Nothing in this Annex creates, implies, expands, or pre-satisfies execution authority.

The core principle remains unchanged: **the system prepares, the human authorizes, and the system records cryptographic proof.**

VIII.1 Purpose, Motivation, and Architectural Principle

VIII.1.1 The Lifecycle Problem of Monolithic Infographics (Informative)

Information graphics are commonly produced and consumed as monolithic raster images. Whether created manually or synthesized by a generative image model, the final artifact fuses text, layout, icons, backgrounds, charts, and illustrations into one pixel surface. This form is convenient as a universal display object, but it carries avoidable costs across the complete lifecycle.

Generation cost. A complete image is synthesized even where most visual content consists of recurring and already-solved elements: containers, arrows, headings, color fields, icons, diagram primitives, typography, and standard illustrations. Full-image generation therefore spends model inference, accelerator time, memory bandwidth, and energy on recreating visual structures that could have been retrieved and reused.

Iteration cost. A correction to one value, label, icon, or section can require regeneration of the complete image. The cost of the smallest change approaches the cost of the whole artifact, and revisions may unintentionally alter regions that were already correct.

Integration cost. A website author receives a static end image rather than structured, responsive content. If the same visual structure is to become semantic HTML, accessible content, or a responsive component, substantial manual reconstruction may be required.

Transfer cost. A raster infographic may occupy hundreds of kilobytes or several megabytes. It is downloaded from the generation service, uploaded to an origin server, transferred through networks, and delivered to every uncached viewer. At large scale, small per-artifact inefficiencies accumulate across origin bandwidth, network transit, storage, caches, backups, and client retrieval.

Adaptation cost. A raster artifact is fixed to one aspect ratio and pixel geometry. Different device classes require scaling, cropping, or separate regeneration. Text cannot naturally reflow, and a mobile device may receive pixels intended for a desktop surface.

Semantic loss. Text fused into pixels is not natively selectable, searchable, machine-translatable, indexable, or consumable by assistive technology. Those properties can only be approximated through later reconstruction.

Caching loss. The same icon, diagram primitive, or background is repeatedly embedded inside unrelated images. The client cannot retrieve an element once and reuse it across artifacts because no element boundary remains.

Syndication loss. A raster artifact syndicated to a third-party site is an opaque file: the receiving site cannot verify its integrity against a declared source, cannot update one value without receiving a complete replacement file, and cannot preserve the provenance of individual elements. Republication severs the audit chain.

The defect is therefore not limited to slow image generation. It is a lifecycle defect extending from reasoning and creation through revision, export, integration, syndication, transfer, rendering, reuse, accessibility, and later modification.

VIII.1.10 Outlook: Handshake-Conveyed Computer-Use Execution (Informative; subject to evaluation). A natural extension of the hybrid principle is expressly contemplated but not specified normatively in this Annex: the cryptographic hand-over of computer-use actions to the operator's local machine through the public handshake. In that contemplated stage, an interaction surface of this Annex — an edition, an integrated block, or an overlay smart action — would not merely present a prepared action but convey it, under a handshake-committed contract, to a local execution authority on the operator's PC, where the action executes within pre-declared execution chains: the available chains are fixed in the handshake in advance, the model acts as a router selecting among declared chains and binding parameters within declared constraints, and execution, witnessing, and proof follow the governed execution discipline specified elsewhere in the Canon. Nothing in this outlook weakens the boundaries of this Annex: composition remains preparation and presentation; suggestion never pre-satisfies authorization; and no computer-use capability arises from any artifact of this Annex alone. The security properties of such hand-over — including the containment of the local execution authority, the integrity of the chain declarations, and the interaction with per-instance consent — require dedicated evaluation before any normative specification; this subsection records the direction, not a guarantee.

VIII.1.2 The Composition Principle (Informative)

This Annex takes the opposite construction:

> **A visual information artifact is composed from reusable, annotated, optimized elements. It is not required to be synthesized as one indivisible image.**

A reasoning system determines the information architecture, selects suitable elements, arranges them within a constrained layout grammar, and writes the textual content. A retrieval and recommendation layer draws recurring visual substance from an Element Library. Only an element that the library cannot adequately supply requires new visual synthesis.

The approach separates four kinds of work:

1. **semantic reasoning —** what the artifact must communicate;
2. **composition reasoning —** which elements and layout best communicate it;
3. **asset provision —** retrieval of existing elements and bounded generation of genuinely missing elements;
4. **deterministic delivery —** responsive rendering and derivation of required output formats.

This separation permits model capacity otherwise spent on full-image pixel synthesis to be allocated to stronger reasoning, factual verification, responsive layout, accessibility, and quality control.

VIII.1.3 Parallel Micro-Asset Generation (Informative)

Parallelizable element production. The decomposition of the artifact creates an additional compute advantage. Once semantic and composition reasoning have determined the layout, element roles, target geometry, style constraints, and placement, independently addressable visual elements — such as icons, pictograms, chart components, connectors, and small illustrations — may be produced concurrently. A high-capability cloud model may perform the semantic and composition reasoning and emit bounded asset descriptions, while smaller specialized or local image-generation models generate the individual assets in parallel. Because these assets are typically small and independently replaceable, each can be generated, validated, retried, cached, and reused without regenerating the complete artifact. A monolithic infographic remains one coupled generation task even where its internal model operations are parallelized; Hybrid Composition instead exposes task-level parallelism across separate assets. This permits heterogeneous compute allocation, reduces dependence on the largest cloud image-generation providers, isolates failures to individual elements, and can shorten the time required to complete the visual artifact.

VIII.1.4 General Applicability Beyond WR, and Defensive Publication (Informative)

The architecture is not limited to WR Desk™. A general-purpose LLM platform, image-generation provider, content-management system, or design tool may implement the same pattern:

1. generate a structured composition plan;

2. retrieve optimized components from a continuously improved library;
3. generate only missing components;
4. assemble a responsive HTML artifact;
5. derive images or documents from that same composition;
6. cache and reuse components — including complete layouts — across artifacts;
7. optimize the process through measured feedback loops;
8. syndicate the resulting content blocks to third-party pages under a verifiable contract.

In such systems, bounded real-time generation of missing elements may be a normal implementation choice. The WR-specific restrictions in this Annex are governance choices for a privacy- and determinism-oriented environment; they are not assertions that request-time generation is technically invalid in every system.

Defensive publication and illustrative embodiment. This Annex is additionally intended to serve as a defensive publication of a general architectural pattern for real-time visual information delivery, independent of any single product, domain, or publisher class. In the preferred form of this pattern, a reasoning model — including a locally operated model fine-tuned or adapted for responsive visual composition — authors the structured composition plan itself: visual hierarchy, grid structure, element placement, breakpoint behavior, stacking, reordering, scaling, and omission priorities are model-authored outputs expressed as a schema-constrained Composition Description, not selections from a fixed catalog of pre-authored page designs. The Composition Engine then deterministically renders that validated description into passive responsive HTML/CSS as the primary Delivery Form. WebP serves as the default raster format throughout the pattern, as the preferred picture format for web delivery: individually generated visual elements — icons, pictograms, and other micro-assets produced by smaller local models — are generated and stored as WebP, and complete raster derivations of the composed artifact are likewise emitted as WebP for export and distribution. Bounded generation by such local models remains confined to genuinely missing visual elements.

The disclosed pattern covers a spectrum of layout-authoring implementations by local models. At one end, the local model emits only the Composition Description, and all HTML/CSS is produced deterministically by the Composition Engine. At the other end, a locally operated model fine-tuned for responsive layout — including a LoRA-class adapter specialization — directly generates the responsive HTML/CSS representation of the artifact itself; in that variant, the generated markup remains within the pinned passive HTML/CSS subset, passes the same structural, responsive, accessibility, and security validation as engine-rendered output, and is normalized into a canonical Composition Description with resolvable element provenance, so that derivation, incremental revision, caching, recycling, and audit continue to operate on one structured source. Both ends of the spectrum, and hybrid intermediate forms — for example, model-generated layout skeletons refined by a deterministic constraint solver, or a recycled prior layout corrected by a bounded model-authored Layout Delta (VIII.6.8) — are expressly contemplated. The defining property is not which component emits the markup, but that a locally operated, layout-specialized model determines the responsive visual arrangement, and that no artifact is displayed or exported without deterministic validation.

The facets of the pattern — semantic reasoning over the information, composition and responsive layout authoring, textual content generation, and micro-asset generation such as icons and pictograms — are separable tasks and may be distributed across a multi-faceted, orchestrated process combining local and cloud models. An orchestration layer may assign each facet to a local model, a cloud model, or a cooperating pair of both, according to capability, cost, privacy sensitivity, latency budget, and reproducibility requirements, and may run the resulting tasks in parallel: once the composition plan has fixed element roles, target geometry, and style constraints, independently addressable assets and independent facets proceed simultaneously — for example, icon and micro-asset generation by local image models concurrently with textual reasoning by a cloud or local LLM, while a layout-specialized local model authors or refines the responsive structure. Every assignment in this matrix is expressly contemplated, from fully local operation to cloud-assisted variants, provided that in WR deployments all external interaction remains subject to the Cloud Abstraction Boundary and the hybrid compute split of VIII.1.6. The purpose of this hybrid combination is explicit and central to the disclosure: it joins the strengths of both sides — maximal task-level parallelization across many small concurrent generation tasks, best-available reasoning quality from high-capability cloud models, and low-latency, low-cost, controllable generation from local models — while preserving privacy by construction. Cloud models operate on public, abstracted, or taxonomy-level information only, and their contributions may be prefetched during preparation; local models then combine those prefetched cloud results with locally held private data, which never leaves the deployment. Public knowledge and private context are thereby processed in combination — on a website, in public delivery, or in mixed scenarios — without routing private data through any cloud provider and without an intermediary between the operator and the resulting artifact: the power of cloud-scale AI is fully exploited, yet the private half of the computation remains local, so that strong AI capability and data privacy are obtained together rather than traded against each other.

Chart and diagram production within the pattern likewise combines deterministic chart-generation tools with fine-tuned adapters: a LoRA-class or similarly specialized adapter translates provenance-bound data into a precise, schema-valid chart specification — series, axes, scales, units, and labels — which a governed chart renderer then materializes deterministically; the adapter authors the specification and never invents the values, which bind from declared sources. The same locality argument extends to richer interactive document components: spreadsheets, table calculations, and office-format documents — for example LibreOffice documents — may be generated locally and embedded into the composed content in interactive form, interpreted by trusted local applications or the host runtime. Because both generation and interpretation occur entirely within the local governed deployment, such components can be offered interactively without transporting executable content in the artifact itself; exported Delivery Forms carry them as static renditions or as declarative references, preserving the passivity of the primary form.

Frame-structured newspaper presentation. The pattern is explicitly directed at upgrading chat-based AI systems as a class. Contemporary chat-based assistants deliver their answers in essentially two forms: as generated text, or as generated monolithic images — with all of the lifecycle defects described in VIII.1.1. TimesDesk™ is the attempt to merge a hybrid of locally operated AI and cloud capability into one interaction surface that transcends both forms: it strongly extends the chat experience visually, interactively, and with governed AI automation, while achieving performance gains through massive parallelization — many small composition, layout, text, and micro-asset tasks executed concurrently across heterogeneous local and cloud compute rather than one serial generation task. In the contemplated interaction model, the user converses as with a chatbot, but the response surface is a composed, provenance-bound visual edition rather than a plain-text transcript. The edition is structured as **Frames**: independently addressable sections whose visual arrangement follows a newspaper-style presentation grammar, in which the most relevant items are rendered with the greatest visual salience — lead position, area, and typographic weight — exactly as an editorial front page prioritizes its stories. The relevance ranking that drives this arrangement is computed by a **Bayesian priority computation coupled to a local or cloud reasoning model**: the model contributes structured relevance assessments as evidence, the Bayesian computation produces a reproducible posterior ranking from pinned inputs, automated research-inspired optimization loops refine the ranking parameters outside the presentation path, and explicit operator feedback — approval and disapproval signals such as thumbs up or thumbs down, together with free-text descriptions parsed into structured evidence — flows into the same computation. Ranking determines what matters; a deterministic priority-to-salience mapping and the frame grammar determine where and at what visual weight it appears. One illustrative embodiment is a system such as TimesDesk™: a real-time visual information generator returning graphically enriched, newspaper-style compositions assembled in multiple editorial frames or sections — summaries, comparisons, charts, callouts, and media references presented as one coherent interactive real-time edition. Each frame or sub-element remains independently responsive, independently exportable — either as a WebP graphic in multiple declared formats or as a responsive HTML fragment — and video material may be incorporated as attested or declared media references within the composed artifact. Interaction with linked external pages, and with the deterministic AI automation offers those pages may present, is contractually fixed in advance and exactly scoped through the public handshake — the visitor sees and agrees the declared capability scope before anything runs — and the rendering and execution pipeline of this Annex enforces deterministic execution: nothing is displayed without validation, and nothing executes outside the agreed, governed path.

Privacy through obfuscation, masking, and distribution obfuscation. In addition to the Cloud Abstraction Boundary and the local/cloud split described above, the pattern contemplates three request-shaping techniques that keep the interactive presentation of the chat private and impede profiling of the operator. *Obfuscation* transforms outbound requests into forms that conceal the operator's concrete intent while remaining useful for the receiving model. *Masking* removes or replaces identifying, private, or correlatable values with placeholders or taxonomy-level abstractions before any request leaves the deployment, with mandatory local re-instantiation of the returned generic result. *Distribution obfuscation* deliberately partitions the work of one interaction into sub-tasks and distributes them across several different LLM or generation providers, so that no single external provider observes more than an isolated, decontextualized fragment of the interaction — no provider receives the complete question, the complete edition, or the complete behavioral trace. Because, in addition, the recommender and ranking systems of the pattern — element recommendation, Mini App selection, the Bayesian Priority Engine, and feedback processing — are operated locally, behavioral profiles never form on any external side: profiling of the operator by a cloud provider is thereby rendered extremely unlikely by construction, not by policy promise. These techniques compose with, and do not replace, the disclosure rules of Annex III and Annex VI.

WR Ads (informative cross-reference). WR Ads, specified separately in the Canon, may optionally be used within this pattern to reduce or offset the token costs of connected external API services used for cloud-assisted facets. Participation is strictly voluntary and separately consented; declining WR Ads never degrades the composition chain. Consistent with the rest of the pattern, WR Ads operate data-locally: ad selection and relevance estimation are performed by the locally operated recommender systems, behavioral and interaction data do not leave the local device, and no external ad or cloud provider receives the operator's interaction history — which is precisely why profiling by cloud providers remains extremely unlikely even where WR Ads are enabled.

Layout Recycling. Because the layout of every composed edition is itself a structured, model-authored artifact, the pattern includes the systematic reuse of previously authored layouts. Every composed frame arrangement is characterized by a **Priority Composition Signature**: a compact, deterministic characterization of the priority structure of the composition — frame roles, ranking weights, geometry classes, and device coverage. When a new composition exhibits a priority structure closely similar to a previously validated layout, the prior layout is recycled: its structure is retained, its content slots are re-bound to the new provenance-bound content, and the artifact is revalidated. Where the new priority structure deviates slightly from the recycled layout, the deviation is corrected in real time by the locally operated, layout-specialized model — fine-tuned or LoRA-adapted for responsive HTML — as a bounded, grammar-constrained **Layout Delta** over the validated recycled structure, followed by mandatory revalidation before display. Recycled layouts are stored in the Element Library as governed assets carrying their Priority Composition Signature, provenance lineage, and an **Optimization Score**. Load-time optimization and responsive-design optimization of stored layouts are performed by automated research-inspired loops exclusively in idle windows, never in the real-time presentation path; the optimized variants are stored back into the library together with their priority pattern and score.

WR Anchor integration. Beyond export as files, the pattern includes contract-bound integration of composed content into third-party pages. A page participating in the WR system declares one or more **WR Anchors**: pre-defined, identified integration positions at which external composed content may appear. Export of a complete edition or of individual frames to such a page proceeds deterministically through the **public handshake**: the receiving operator sees the declared scope and integrates the content with a single connect-and-integrate action at the previously defined anchor position. The exported code and every generated element — HTML, CSS, and image assets — are part of the handshake contract: each asset is committed as a leaf of the handshake's Merkle manifest, so that the integrated content is cryptographically provable against the contract root. The image and asset files themselves are hosted locally on the receiving server — a dedicated asset directory is created on the target for this purpose, a step that a later architecture stage may automate — so that the receiving site serves the content under its own identity, with no runtime dependency on the exporting deployment and no hotlinked external resources, while integrity remains verifiable against the committed manifest. Updates to integrated content follow a versioned update path: content is hash-pinned at integration; a change is published only as an explicit new contract version referencing the prior root, surfaced to the receiving operator under the update policy agreed in the contract; silent replacement does not occur. Where the surrounding page participates in the WR system, embedded WR Links within the integrated block enable deterministic, public-handshake-based interaction, as specified below.

Beyond static visual composition, the pattern incorporates three interactive extensions: the Augmented Overlay, Mini Apps, and WR Links. The Augmented Overlay is an operator-toggleable interaction layer providing pointer-driven real-time support on the composed artifact; it surfaces prepared smart actions which, after explicit, recorded operator consent, execute through the governed authorization path (PoAE™) — suggestion never pre-satisfies authorization, but authorized execution is an intended capability, not an exception. The overlay additionally serves as the pre-export editing surface of the pattern: specially marked regions of a composed artifact — declared as Editable Regions in the composition plan — are modifiable through overlay interaction before export, either manually by the operator or with AI assistance by the local models of the pattern, at the granularity of individual text slots, values, elements, or frames; every such edit flows through the ordinary composition machinery, produces a revised canonical composition, and is revalidated before the artifact is exported. Mini Apps are recommender-selected interactive components — calculators, configurators, comparison and filtering widgets, parameter-bound visualizations — assembled declaratively from pre-admitted interaction primitives held in the local Element Library and bound into typed interactive slots of the Composition Description; their interaction behavior is supplied exclusively by the trusted host runtime, no executable code is authored by a model, generated at request time, or transported with the artifact, and the primary Delivery Form remains free of scripts and active content. WR Links are declarative connection references — including WR Code® entry points — embedded directly within the infographic: identifier and invitation, not authority, with activation proceeding exclusively through the governed initiation layer. Together, these extensions make the composed artifact operable rather than merely informative, under the same admission, provenance, and authorization discipline that governs visual elements.

The exportability of responsive HTML content blocks is itself part of the disclosed pattern. Each frame, section, or sub-element of a composed artifact is exportable as a self-contained responsive content block that a publisher may host and embed under its own identity on its own web presence, with embedded WR Links intact; further WR Links may be added directly to an exported block or composition — by the composing system or by the publisher — without altering the block's passive character. Where the surrounding page participates in the WR system, those entry points enable deterministic, public-handshake-based interaction: a visitor who follows a WR Link enters the governed handshake, in which the available automations are declaratively presented and agreed in advance — the visitor sees the declared capability scope and consents to it before anything runs. Execution of an agreed automation then occurs inside the WR system under its deterministic execution and authorization discipline, not inside the exported block, so the content surface and the execution environment remain strictly separated: the page invites, the handshake agrees, the WR system executes. Special WR Links may additionally be pinned as favorites to the Augmented Overlay, where they appear as quick-select entries in the cursor toolbar: the operator attaches a tailored automation — a local automation or one previously agreed with a particular site through the handshake — and can invoke it with a single gesture on the corresponding page, under the same consent, provenance, and deterministic-execution discipline as any other smart action; pinning changes convenience, never authority. Every published infographic thereby becomes a distribution and interaction surface for governed, deterministic automation, allowing the pattern to scale through publisher channels without weakening the passivity, provenance, or authorization model.

First essential combined disclosure. The essential combined disclosure of the base pattern is therefore: conversational interaction whose response surface is a responsive, provenance-bound visual edition whose layout is authored by locally operated, layout-specialized models — whether as a schema-constrained composition plan or as directly generated, validated responsive HTML — produced by an orchestrated, parallelizable local/cloud model process in which reasoning, layout authoring, text generation, and asset generation are independently assignable facets, and in which prefetched cloud reasoning over public information is combined locally with private data that never leaves the deployment — preserving privacy without an intermediary while retaining cloud-scale capability; together with reusable and recommender-selected visual elements and Mini Apps, adapter-driven precise chart generation, locally generated and locally interpreted interactive spreadsheet and document components, Augmented Overlay support with consent-gated smart-action execution and pinned WR Link favorites, embedded WR Links as governed connection entry points, exportable responsive content blocks whose interactions are agreed with visitors through the public handshake and executed deterministically within the WR system, and a multi-form export chain derived from one canonical composition — as a general architectural upgrade path for chat-based AI systems. The privacy techniques of obfuscation, masking, and distribution obfuscation across multiple model providers, the locally operated recommender and ranking systems that prevent external behavioral profiling, and the optional, voluntary, data-local WR Ads mechanism offsetting connected-service token costs are expressly disclosed as parts of this combination.

Second essential combined disclosure. The essential combined disclosure of the editorial, recycling, and integration extension is: a frame-structured, newspaper-style visual edition whose frame arrangement is authored by a locally operated, layout-specialized model and whose visual salience is derived deterministically from a Bayesian priority computation coupled to a local or cloud reasoning model, refined by automated research-inspired optimization loops and by structured operator feedback evidence; each composed arrangement characterized by a Priority Composition Signature; previously validated layouts recycled by signature match, re-bound with new provenance-bound content, and — for near-matches — corrected in real time by a bounded, grammar-constrained, model-authored Layout Delta with mandatory revalidation; recycled layouts stored as governed library assets carrying signature, lineage, and Optimization Score; load-time and responsive optimization of stored layouts performed exclusively in idle windows outside the presentation path; complete editions and individual frames individually responsive and individually exportable as responsive HTML fragments or WebP raster derivations; and integration of such content into third-party WR-connected pages through declared WR Anchors under a public-handshake Integration Contract in which every transferred asset — HTML, CSS, and images — is committed as a leaf of the handshake Merkle manifest, the asset files are hosted locally on the receiving server in a dedicated asset directory under the receiving site's own identity, integrity is verifiable against the contract root, updates occur only as explicit, consent-governed new contract versions referencing the prior root, and the provisioning of the receiving asset directory may itself be automated in a later architecture stage. Every sub-combination of these features with the base pattern of this Annex is expressly contemplated and disclosed.

The hybrid principle as the defining architecture. The local/cloud combination described above is not an implementation convenience of one component; it is the defining principle of the disclosed pattern, and it generalizes beyond infographics to every artifact class this Annex touches. The pattern deliberately splits every generation task along one line: whatever requires broad public knowledge, large-scale reasoning capability, or expensive semantic work is obtained from cloud models — operating exclusively on public, abstracted, or taxonomy-level inputs, and wherever possible prefetched during preparation; whatever touches private data, produces the final artifact, or must be reproducible and auditable is performed by locally operated models and deterministic local machinery. The two halves meet only locally: prefetched cloud contributions are combined with locally held private context inside the deployment, and the finished artifact — an edition, a frame, a chart, a document — exists only there. This split is applied uniformly to every facet (reasoning, layout authoring, text generation, micro-asset generation, ranking, recommendation) and to every artifact class (visual editions, charts, office-format documents, interactive components). The consequence is stated once and holds everywhere: cloud-scale capability and data privacy are obtained together, by construction, because the boundary between them is architectural rather than contractual.

Office artifacts generated locally from handshake-transferred data. The same hybrid split governs the integration of word-processing documents, spreadsheets, and other office-format artifacts with connected external parties. Where a WR-connected page, publisher, or counterpart supplies data intended to become or to populate an office document, that data is handed over through the public handshake as text-pure, schema-validated payloads: structured values, tables, and textual content, committed in the handshake manifest — never as binary document containers, embedded objects, macros, or any other potentially active content. The receiving deployment then generates the office artifact itself, locally and under its own governance: local models author structure and text over the transferred provenance-bound values, and trusted local applications or the host runtime materialize and interpret the resulting document. The document container is therefore always locally synthesized; no foreign office file ever crosses the boundary. This construction removes the classic document-borne attack surface (macro payloads, exploit-bearing containers, parser attacks) by design rather than by scanning: the handshake transfers data, and only data — the artifact is born locally. Conversely, where locally generated office content is exported or integrated at a WR Anchor, the exported Delivery Forms carry it as static renditions or declarative references under VIII.7, preserving the passivity of the primary form.

An audiovisual, not merely visual, response surface. The composed edition is correctly characterized as audiovisual and interactive, in three respects. First, the operator interacts with the surface directly through pointer and cursor: the Augmented Overlay (VIII.9) provides pointer-driven real-time support, annotation, pre-export editing of declared Editable Regions, and consent-gated smart actions, so the artifact is an operable surface rather than a static picture. Second, speech participates as a locally transcribed input modality under VIII.9.4. Third, video participates as content: suitable external video material — for example a platform-hosted video such as a YouTube video matching the topic of a frame — may be incorporated as an attested or declared media reference with time range under VIII.3.7, retrieved under the privacy discipline of Annex VI and preferably during the preparation window. The passivity guarantees of VIII.7 are unaffected: interactivity is supplied by the trusted host runtime and the governed overlay, never by the exported artifact, and referenced media remain references, not embedded active content.

VIII.1.5 Scope and Delimitation (Normative)

The normative provisions bind the WR presentation chain: the preparation phases of Annex VI, the composition sources of VIII.6, the Delivery Forms of VIII.7, and the Presentation Surfaces of WR Desk™ and paired devices.

This Annex governs:

- how visual information artifacts are planned and represented;
- which visual elements may be used and under what provenance;
- how missing elements may be generated;
- how frame priority is computed and mapped to visual salience;
- how layouts are recycled, corrected, and stored;
- how operator-selected generation profiles alter guarantees;
- how responsive HTML and raster outputs are derived;
- how composed content is integrated into third-party pages through WR Anchors and updated thereafter;
- how efficiency and quality are measured;
- how visual artifacts are presented, audited, and degraded.

This Annex does not govern task execution, transport authorization, or the semantics of external actions. A composed artifact may present an action or recommendation, but it never grants authority to execute it.

VIII.1.6 The Hybrid Compute Split (Normative)

Composition work SHALL be placed according to its cost, privacy sensitivity, and reproducibility requirements.

- **Preparation phase:** broad semantic exploration, candidate generation, exemplar retrieval, library optimization, asset generation, responsive testing, layout optimization, and expensive quality evaluation.
- **Presentation phase:** deterministic branch evaluation, scored retrieval, bounded selection, recycled-layout matching, slot binding, permitted refinement, bounded Layout Delta correction, rendering, and — only where an authorized generation profile permits it — bounded generation of missing visual elements.
- **Idle windows:** automated research-inspired optimization of stored layouts, ranking parameters, and delivery characteristics, under VIII.11.
- **External services:** only under the Cloud Abstraction Boundary of Annex III §7 and the retrieval-privacy rules of Annex VI §VI.7. Concrete WR Graph™ content, operational knowledge, and private composition-instance data SHALL NOT cross that boundary unless a separately defined and explicitly authorized disclosure path applies.

External reasoning is an accelerant, never a dependency. The complete composition chain SHALL remain operable in a local-only conforming mode.

VIII.1.7 The Default Inversion Rule (Normative)

Under the default WR Deterministic Profile:

> **Neural work that authors composition structure or visual assets SHALL occur in preparation, where its output is validated and captured as provenance-bound artifacts. The presentation path SHALL be deterministic except for the four bounded exceptions enumerated below.**

The four permitted presentation-time exceptions are, exhaustively:

1. **bounded selection** under VIII.6.3;
2. **grammar-constrained textual generation** over provenance-bound facts under VIII.4.3;
3. **the staged composition fallback** defined in VIII.6.6;
4. **the bounded Layout Delta** over a validated Recycled Layout Asset under VIII.6.8, subject to mandatory revalidation before display.

The rule is an inversion of ordinary on-demand graphics generation. Information carries correctness obligations; presentation is therefore not the default place for unbounded creative synthesis.

As a corollary, optimization work — including load-time optimization and responsive-design optimization of stored layouts — SHALL NOT execute in the presentation path. It executes exclusively in preparation phases and idle windows under VIII.11.

The Inversion Rule is a default WR assurance rule, not a universal prohibition. It MAY be relaxed only through the explicit generation profiles of VIII.1.9 and VIII.5.4.

VIII.1.8 Composition, Not Whole-Image Synthesis (Normative)

Every displayed visual component SHALL originate from one of:

1. the host asset base;
2. an admitted Publisher Composition Library;
3. a Derived Composition Asset prepared locally, including Recycled Layout Assets;
4. a Live-Derived Composition Asset generated under an authorized profile;
5. operator-local data bound into typed slots.

A conforming implementation SHALL prefer retrieval and reuse over synthesis whenever an admitted element — or an admitted recycled layout — satisfies the declared need above the applicable quality threshold.

Full-image pixel synthesis MAY be offered as a non-primary fallback or export technique, but it SHALL NOT be treated as equivalent to Hybrid Composition unless the resulting artifact is reconstructed into a valid CD with resolvable element provenance and passes the same validation requirements.

Textual content MAY be generated at presentation time. Every factual value within generated text SHALL bind from provenance-bound sources: narrative may be generated; facts are bound.

VIII.1.9 Generation Profiles and Operator Choice (Normative)

The WR runtime SHALL expose at least the following profiles:

A. WR Deterministic Profile — default

- presentation-time visual synthesis disabled;
- only admitted, reconstructed, hash-bound assets displayed;
- missing decorative elements omitted, replaced by an admitted neutral element, or queued for preparation;
- exact reproduction of displayed artifacts required;
- the bounded Layout Delta of VIII.6.8 permitted, hash-retained, and revalidated;
- no additional external interaction caused by rendering.

B. WR Local Live Generation Profile — optional

- bounded local generation of missing visual elements permitted;
- no composition data leaves the deployment;
- generated assets SHALL pass validation before display;
- generated assets carry `live-derived-local:` provenance;

- exact byte-for-byte regeneration is not guaranteed unless the complete generation state is retained, but the displayed output itself SHALL be hash-retained and reproducible as a rendered artifact;
- activation SHALL be scoped and revocable.

C. WR External Live Generation Profile — optional, highest disclosure class

- bounded use of an external generation service permitted;
- activation requires an informed operator decision under Annex III and the disclosure scope applicable to the request;
- the information dialog SHALL identify what data or abstractions may leave the deployment, the receiving class of service, expected cost, latency, provenance consequences, and reduced reproducibility;
- generated assets SHALL be reconstructed under Annex V before display and carry `live-derived-external:` provenance;
- activation SHALL be separately controllable from local live generation.

The operator MAY authorize a profile for one element, one artifact, one session, one context region, or a persistent policy scope. Persistent authorization SHALL remain discoverable and revocable. Viewing, waiting, or continuing a conversation SHALL NOT constitute authorization.

The system SHALL explain the guarantees changed by the selected profile in concrete terms: privacy, external disclosure, provenance, reproducibility, quality-floor status, latency, and cost. The dialog SHALL not use a generic warning as a substitute for an informed choice.

VIII.2 Definitions

Terms defined in Annexes II–VII and IX apply unchanged. This Annex adds:

Composed Artifact. A visual information artifact produced from a validated Composition Description over hash-addressed elements and provenance-bound data. The canonical artifact is structured; one or more rendered Delivery Forms may be derived from it.

Composition Description (CD). The declarative, schema-constrained specification of a Composed Artifact: layout, frames, element references, slots, responsive behavior, presentation policy, provenance, and output metadata. The CD is the unit of validation, staging, derivation, and audit. All composition state — including the association of content to layout positions — lives inside the CD; no separately stored specification artifact controls composition.

Composition Engine (CE). The governed renderer transforming a valid CD into one or more Delivery Forms. Its identity includes version, content hash, rendering subset, constraint-solver version, and supported Render Profiles.

Composition Grammar. The permitted composition space: layout structures, frame structures, slot types, responsive constraints, equivalence classes, transformations, and validation rules.

Frame. An independently addressable section of a Composed Artifact, declared in the CD, carrying its own role, salience weight, responsive constraints, provenance set, and export capability. A frame is individually responsive and individually derivable as a responsive HTML fragment or a raster derivation.

Frame Grammar. The newspaper-style presentation grammar governing frame arrangement: lead positions, area classes, typographic weight classes, sectioning, and the deterministic mapping from priority rank to visual salience.

Bayesian Priority Engine. The local component computing a reproducible posterior priority ranking over candidate frames and their contained composed graphics from pinned priors, structured model-contributed relevance evidence, and accumulated local evidence including operator feedback.

Priority Evidence. Structured, locally recorded evidence entering the Bayesian Priority Engine, including declared preparation relevance, model-contributed relevance assessments, engagement telemetry, and explicit operator feedback (approval/disapproval signals and parsed free-text descriptions).

Priority Composition Signature (PCS). A deterministic, compact characterization of the priority structure of a composed arrangement, computed under VIII.6.8.1 from frame roles, normalized ranking weights, geometry classes, and device coverage, used to index, match, and recycle layouts.

Recycled Layout Asset. A previously validated composed layout, stripped of instance content, stored in the Element Library as a Derived Composition Asset of layout class, carrying its PCS, provenance lineage to its originating CD, validation record, and Optimization Score.

Layout Delta. A bounded, grammar-constrained set of structural modifications authored by the Layout Model over a Recycled Layout Asset to reconcile a near-match between a new priority structure and the recycled structure. A Layout Delta is recorded, hash-retained, and subject to mandatory revalidation before display.

Layout Model. The locally operated, layout-specialized model — fine-tuned or LoRA-adapted for responsive HTML/CSS and CD authoring — that authors composition structure, responsive HTML within the pinned passive subset, and Layout Deltas.

Optimization Score. A versioned, reproducible quality/efficiency score attached to a stored layout or element, produced by the measured optimization loops of VIII.11, covering at minimum load-time characteristics, responsive stability, and quality-floor conformance.

Idle Optimization Window. A preparation or idle period in which optimization loops may execute. The presentation path is never an Idle Optimization Window.

WR Anchor. A pre-defined, identified integration position declared by a WR-connected page, at which externally composed content may be integrated under an Integration Contract. An anchor declares an identifier, accepted content classes, geometry class, style-isolation requirements, and its Anchor Asset Directory location.

Anchor Declaration. The machine-readable declaration of a WR Anchor by the receiving page or its operator, exchanged through the initiation layer (Annex IX) or the handshake capability declaration (Annex VII).

Integration Contract. The public-handshake contract governing integration of composed content at a WR Anchor: parties, anchor identifier, content block identity, the Merkle manifest root committing every transferred asset, update policy, licensing and attribution terms, and revocation terms.

Anchor Asset Directory. The dedicated directory on the receiving server in which the transferred asset files — images, CSS, and HTML fragments — are hosted locally under the receiving site's own identity.

Content Update Notice. The explicit, contract-versioned notice by which a change to integrated content is offered to the receiving operator under the agreed update policy.

Element Library. The total set of reusable composition elements available to the CE across host, Acquired, Derived (including Recycled Layout Assets), and authorized Live-Derived provenance classes.

Publisher Composition Library (PCL). Publisher-produced assets, grammars, exemplars, domain presentation policy, optimization metadata, and media references published and admitted under Annex VII and PoAC™.

Derived Composition Asset. A locally prepared and validated reusable visual or structural element carrying `derived:` provenance.

Live-Derived Composition Asset. A missing element generated during an active request under an operator-authorized generation profile, validated before display, hash-retained, and carrying `live-derived-local:` or `live-derived-external:` provenance.

Element Annotation Record. Machine-readable metadata describing an element's semantics, style, geometry, responsive behavior, accessibility properties, provenance, quality metrics, Optimization Score where applicable, licensing, and compatibility constraints.

Gap Detection. Deterministic or bounded-model identification that no admitted library element meets a declared need above the applicable fitness and quality thresholds.

Delivery Forms. Outputs derived from one canonical CD and its asset set. Responsive HTML is the primary form; raster snapshots and document forms are secondary derivations.

Responsive HTML Package. A passive, self-contained HTML/CSS representation plus its hash-addressed asset manifest, responsive rules, accessibility metadata, and integrity manifest.

Raster Snapshot. A pixel rendering of a CD or of an individual frame produced by a governed Rasterizer under a declared Render Profile.

Render Profile. A versioned set of parameters governing a rendering derivation, including viewport, device pixel ratio, locale, color space, font set, background, output dimensions, format, encoder, compression, and quality settings.

Delivery Optimizer. The local deterministic component selecting an available Delivery Form for a target environment under quality, capability, privacy, transfer, and rendering-cost constraints.

Candidate Set. Event-bound, pre-validated CDs produced during a Preparation Run and selected by Branch Conditions.

Branch Condition. A deterministic predicate over locally available runtime data selecting a prepared candidate without inference.

Session Composition Pool. Session-scoped pre-generated and pre-validated CDs serving Real-Time Support through deterministic fitness matching.

Fitness Function. A deterministic, reproducible scoring algorithm ranking elements or composition candidates against a structured need descriptor.

Refinement Vocabulary. The closed set of deterministic transformations a grammar permits on an already validated candidate.

Prepared Edition. The editorial visual presentation form of the Morning Digest and Preparation Briefs.

Augmented Overlay. The operator-toggleable interaction layer rendered over passive composed artifacts.

Editable Region. A specially marked slot, element, or frame of a Composed Artifact declared in the CD as modifiable before export through Augmented Overlay interaction, manually by the operator or with AI assistance, under VIII.9.5.

Trigger provenance. `declared` for confirmed preparation and `inferred` for Preparation Queue hypotheses.

Quality Floor. The minimum acceptable correctness, legibility, accessibility, responsive stability, and visual-integrity thresholds below which an element or composition SHALL not be presented at full visual authority.

Lifecycle Efficiency Record. The measurement record comparing generation, revision, package size, transfer, cache reuse, rendering, and quality against a declared baseline.

VIII.3 The Element Library and Its Optimization Lifecycle

VIII.3.1 Element Classes and Annotation (Normative)

The Element Library MAY contain:

- layout containers and section structures;
- Recycled Layout Assets and frame arrangements;
- backgrounds, themes, and design tokens;
- icons, pictograms, and connectors;
- vector and raster illustrations;
- chart and diagram primitives;
- typographic resources;
- responsive layout grammars and frame grammars;
- accessibility metadata and alternate textual descriptions;
- export and Render Profile compatibility declarations.

Every element SHALL be content-hash addressed and SHALL carry an Element Annotation Record containing at least:

1. semantic tags and domain association;
2. element class and declared use;
3. intrinsic geometry, aspect ratio, and permissible transformations;
4. supported responsive size classes and reflow behavior;
5. style family, color semantics, and theme compatibility;
6. accessibility properties, including alternate description where applicable;
7. provenance and admission state;
8. quality metrics, Optimization Score where applicable, and evaluation history;
9. licensing and export rights;
10. supported Delivery Forms;
11. usage, acceptance, rejection, and performance evidence;
12. for Recycled Layout Assets: the Priority Composition Signature and originating-CD lineage.

Annotations inform scoring; they do not themselves grant authority.

VIII.3.2 Provenance Classes (Normative)

Library elements exist in four provenance classes:

1. **host** — local runtime assets;
2. **Acquired** — admitted PCL artifacts;
3. **Derived** — locally prepared reusable assets, including Recycled Layout Assets;
4. **Live-Derived** — request-time assets admitted for the current authorized scope and optionally proposed for later persistent admission.

All displayed elements SHALL be resolvable by hash. No provenance class carries elevated factual authority.

VIII.3.3 PCL Artifact Model, Distribution, and Admission (Normative)

A PCL is a set of publisher artifacts under Annex VII. Every artifact SHALL carry a signed manifest, be committed under the publisher's Manifest Root, and pass the Annex VII verification chain before staging.

Distribution follows the public repository or Private Distribution Path. Public-offering elements SHALL be publicly inspectable and auditable. Non-public elements remain hash-committed under the publisher's root and follow the evidentiary-reveal and attestation rules of Annex VII.

PCL admission is Class 3. Until admitted, its artifacts are unavailable to composition.

Media-class PCL assets SHALL traverse Annex V reconstruction. The CE composes only Reconstructed Artifacts. Text-pure structural and vector assets remain structured artifacts subject to schema and safety validation.

VIII.3.4 PCL Contents (Normative)

A PCL MAY contain:

- visual assets;
- layout grammars and frame grammars;
- responsive breakpoints and container rules;
- curated data-pattern-to-CD exemplars;
- domain presentation policy;
- media references;
- typography and localization policies;
- accessibility requirements;
- Render Profile recommendations;
- quality-evaluation fixtures;
- element-equivalence classes for deterministic refinement.

VIII.3.5 Constraining the Composition Space (Normative)

A publisher SHALL constrain composition through grammars wherever its PCL governs a domain. Free-form placement outside the applicable grammar is non-conformant.

Publishers SHOULD provide exemplars for recurring information patterns. Exemplar retrieval followed by slot binding and bounded adaptation SHOULD be preferred to unconstrained generation.

Where parametric composition competence is supplied through a LoRA-class adapter, Annex IV applies unchanged. Grammars and exemplars are the cheaper and more inspectable rungs and SHOULD be exhausted before adapter acquisition is proposed.

VIII.3.6 Domain Presentation Policy (Normative)

A domain presentation policy SHALL declare, where applicable:

- units, precision, and rounding;
- date, time, number, and locale conventions;
- reserved color meanings;
- source, timestamp, and confidence display requirements;
- chart-axis and scale rules;
- prohibited or misleading encodings;
- accessibility and contrast minima;
- mandatory context for values and forecasts;
- permitted transformations across Render Profiles.

The policy is applied deterministically at slot binding and validation. It never executes.

VIII.3.7 Media References (Normative)

Video and audio SHALL be integrated by attested reference and time range. Reconstruction and display follow Annex V. Retrieval SHALL comply with Annex VI privacy rules and SHOULD occur during the preparation window rather than in a correlatable presentation-time request.

VIII.3.8 Library Retrieval and Fitness Scoring (Normative)

Element retrieval MAY combine semantic embeddings, rules, Bayesian ranking, recommender systems, preference evidence, and learned quality estimates. The final admitted-element ranking used for deterministic presentation SHALL be reproducible from pinned inputs and scoring components.

The Fitness Function for elements SHOULD account for:

- semantic fit;
- style and domain compatibility;
- geometry and responsive fit;
- accessibility;
- quality score and Optimization Score;
- provenance preference;
- expected package-size contribution;
- cache availability;
- licensing and export compatibility;
- historical operator feedback.

A highest-ranked element below the minimum threshold is a miss, not an acceptable match.

VIII.3.9 Gap Detection and Missing-Element Generation (Normative)

Gap Detection SHALL occur only after admitted retrieval has failed to satisfy the declared need above threshold.

A gap record SHALL identify:

1. the required semantic role;
2. geometry and responsive constraints;
3. applicable style and domain rules;
4. the reason retrieved candidates failed;
5. whether the element is informational, structural, or decorative;
6. the selected generation profile;
7. the quality and validation tests required before use.

Under the Deterministic Profile, the gap SHALL be resolved by omission, an admitted neutral substitute, degradation, or preparation queueing.

Under an authorized live-generation profile, only the missing element SHOULD be synthesized. Regeneration of the complete artifact is non-preferred unless the composition structure itself is invalid or no partial path exists.

VIII.3.10 Library Lifecycle and Optimization Loops (Normative)

The Element Library is a governed learning system, not a static asset folder.

The Night Pipeline and other Idle Optimization Windows MAY perform:

- coverage analysis over recurring composition gaps;
- element de-duplication;
- quality re-evaluation;
- annotation enrichment;
- responsive compatibility testing;
- extraction, generalization, and staging of Recycled Layout Assets from validated CDs;
- load-time and responsive-design optimization of stored layouts, with Optimization Score assignment;
- cache-value estimation;
- format and compression optimization;
- proposal of new assets, grammars, exemplars, or equivalence classes;
- controlled benchmark experiments inspired by automated research loops.

Optimization SHALL be quality-constrained. No reduction in byte size, latency, or compute is acceptable if the applicable Quality Floor is violated.

Evidence accumulation is Class 1; deterministic re-ranking is Class 2; generation and staging of local Derived assets — including Recycled Layout Assets — from admitted content is Class 2; acquisition of external assets is Class 3; persistent promotion of Live-Derived assets is governed by their source and disclosure path under VIII.5.4.

VIII.4 The Composition Description and Reasoning Outputs

VIII.4.1 The Three Reasoning Outputs (Normative)

Model reasoning contributes exactly three primary output classes:

1. **element selection** — references to elements or declared gaps;
2. **arrangement** — binding those elements into a grammar-permitted structure, including frame structure and, where applicable, a Layout Delta over a recycled arrangement;
3. **textual content** — headings, labels, explanations, summaries, and narrative over provenance-bound facts.

Pixel-level synthesis is an asset-production activity, not a fourth composition-reasoning output.

VIII.4.2 CD General Requirements (Normative)

A CD SHALL:

- validate against the applicable Composition Grammar;
- be text-pure, referencing assets by identifier and hash;
- be self-contained with respect to its declared asset set;
- declare its frame structure where the artifact is frame-structured;
- declare responsive behavior;
- declare Delivery Forms and supported Render Profiles;
- include provenance, confidence, and generation-profile metadata;
- include, where a layout was recycled, the identity and PCS of the Recycled Layout Asset and the hash of any applied Layout Delta;
- contain no executable content;
- require no inference during deterministic rendering.

Model-produced CDs SHALL use schema-enforced decoding. A CD failing validation SHALL be discarded or explicitly regenerated; it SHALL NOT be silently repaired.

VIII.4.3 Slots and Fact Binding (Normative)

Layout structures SHALL declare typed slots. Slot binding is a pure operation: insert a provenance-bound value, format under domain policy, validate, and render. The slot association is part of the CD itself; it is not a separately stored control artifact.

Slots SHALL distinguish:

- factual values;
- generated narrative;

- citations and source lines;
- media elements;
- optional decorative elements;
- interactive affordance anchors rendered by the surrounding Presentation Surface.

Generated narrative MAY vary. Bound factual values SHALL not be invented by the narrative generator.

VIII.4.4 Responsive Constraints (Normative)

A CD SHALL define responsive behavior through a versioned set of constraints, including:

- grid and flow rules;
- min/max widths;
- breakpoints or container-query conditions;
- element and frame priority and permissible omission;
- text expansion budgets;
- image-source selection;
- permitted stacking and reordering;
- typography scaling;
- fixed relationships that SHALL not be broken.

Responsive adaptation SHALL preserve information meaning and source association. A mobile layout MAY reorder presentation for readability but SHALL not change the logical or evidentiary relationship between elements.

VIII.4.5 CD Output Manifest (Normative)

Every CD SHALL carry or reference an output manifest containing:

- CD version and hash;
- grammar and policy versions;
- asset manifest;
- recycled-layout reference and Layout Delta hash where applicable;
- CE and Rasterizer identities;
- supported Render Profiles;
- default Delivery Form;
- default raster format;
- accessibility metadata;
- generation profile and authorization reference;
- export-rights status;
- derivation and integrity hashes.

VIII.4.6 Frame Structure (Normative)

A frame-structured CD SHALL declare, per frame:

1. a stable frame identifier;
2. the frame role (for example: lead, secondary, comparison, chart, callout, media, source block);
3. the salience weight assigned under VIII.8.2;
4. the frame's own responsive constraints and tested size classes;
5. the frame's element and provenance set;
6. the frame's export capability: responsive HTML fragment, raster derivation under declared Render Profiles, or both;
7. embedded WR Links, if any, as declarative references.

Each frame SHALL be independently valid: it SHALL pass responsive, accessibility, and integrity validation both in the context of the complete edition and as a standalone exported fragment.

Frame arrangement SHALL follow the governing Frame Grammar. The Frame Grammar expresses the newspaper presentation principle: higher-priority content occupies positions and area classes of higher visual salience, deterministically and reproducibly.

VIII.5 Provenance, Generated Assets, and Informed Authorization

VIII.5.1 Trigger Provenance and Confidence (Normative)

Every CD SHALL declare:

- `trigger_provenance` ∈ {`declared`, `inferred`};
- confidence or declared status;
- `preparation_ref` to the applicable Preparation Consent, Preparation Run, or Queue evidence chain.

These fields bind to visual salience under VIII.8.3.

VIII.5.2 Per-Element Provenance (Normative)

Every element SHALL declare one origin:

- `host:<asset-id>@<hash>`;
- `pcl:<publisher-id>:<artifact-id>@<hash>`;
- `derived:<asset-id>@<hash>` — including Recycled Layout Assets, which additionally carry lineage to their originating CD;
- `live-derived-local:<asset-id>@<hash>`;
- `live-derived-external:<provider-class>:<asset-id>@<hash>`;
- `local:<binding>`.

The rendered artifact SHALL preserve the ability to resolve the origin, admission record, generation record, and decision state of every element. Composition influence is never silent.

VIII.5.3 Derived Composition Assets Prepared Locally (Normative)

During preparation and idle periods, the deployment MAY generate reusable assets from admitted content. Such assets SHALL:

1. be produced by a governed generation tool;
2. carry the tool identity, inputs, target context region, and content hash;
3. pass structural, visual, responsive, accessibility, and policy validation;
4. be staged as Class 2 with itemized Morning Digest reporting and rollback;
5. remain deactivatable without falsifying prior Derivation Records;
6. be regenerable from a retained recipe where technically possible.

Rejection deactivates future use and cascades to dependent staged compositions. It does not erase history.

VIII.5.4 Live-Derived Composition Assets (Normative)

A Live-Derived asset MAY be produced only under an active Local Live Generation or External Live Generation authorization.

Before activation, the system SHALL present an information dialog stating:

- whether generation is local or external;
- which information classes are used or disclosed;
- the receiving service class where external;
- the expected effect on determinism and reproducibility;
- the expected latency and cost class;
- the fact that the asset has not passed the same pre-admission history as a library element;
- the validation performed before display;
- the authorization scope and revocation method.

The authorization SHALL be recorded as a Consent Record. The generated asset SHALL not be displayed until it passes all required validation and Annex V reconstruction where applicable.

A Live-Derived asset is scoped by default to the current artifact. Persistent reuse requires a separate admission decision:

- locally generated from admitted content: MAY be proposed as a Derived asset, Class 2;
- externally generated or containing newly acquired external content: Class 3;
- generated under a scope prohibiting persistence: SHALL be discarded after audit-retention obligations are met.

VIII.5.5 Authority Parity (Normative)

Generated does not mean authoritative. Host, PCL, Derived, and Live-Derived elements are subject to the same factual binding, grammar, policy, and salience rules. A generated chart asserts no ground truth of its own; its values bind from declared sources.

VIII.5.6 Authorization Non-Coercion (Normative)

The Deterministic Profile SHALL remain fully functional. The runtime SHALL NOT degrade ordinary usability solely to pressure the operator into enabling live generation.

The operator SHALL be able to:

- decline without losing the textual answer;
- choose local rather than external generation where available;
- authorize only one missing element;
- revoke persistent permission;
- inspect the resulting provenance;
- request deletion where compatible with audit and retention law.

VIII.6 Prepared Composition Sources and Live Presentation

Prepared visual content reaches a Presentation Surface from four sources, in descending order of preparation specificity:

1. Candidate Sets;
2. the Session Composition Pool;
3. Recycled Layout Assets matched by Priority Composition Signature;
4. staged fallback composition, optionally augmented by authorized missing-element generation.

VIII.6.1 Candidate Sets and Branched Preparation (Normative)

For a Declared Event, the Preparation Run SHALL produce parameterized CDs covering plausible branches, each guarded by a deterministic Branch Condition.

Every candidate SHALL:

- validate at preparation time;
- declare pure-bind slots;
- inherit provenance and preparation lineage;
- include responsive behavior and supported Render Profiles;
- contain no unresolved informational gap required for factual correctness.

Candidate Set assembly from admitted content is Class 2. New external Sources or capabilities follow Class 3.

VIII.6.2 Presentation Resolution Path (Normative)

At presentation time, resolution proceeds:

1. evaluate Branch Conditions;
2. where no branch matches, rank Session Composition Pool candidates;
3. where no pool candidate meets threshold, compute the Priority Composition Signature of the required arrangement and match it against stored Recycled Layout Assets under VIII.6.8;
4. perform bounded selection if required;
5. bind current values;
6. apply permitted deterministic refinement and, for a near-match recycled layout, the bounded Layout Delta;
7. detect remaining element gaps;
8. resolve gaps according to the active generation profile;
9. validate;
10. render the selected Delivery Form.

Implementations SHOULD target sub-second time-to-first-correct-display for prepared, pool, and exact-match recycled paths. Telemetry SHALL record branch hit rate, pool hit rate, recycling hit rate (exact and near-match separately), gap rate, selected profile, time-to-first-display, time-to-complete-composition, and Delivery Form.

VIII.6.3 Bounded Selection (Normative)

At presentation time, a local model MAY output only:

1. a prepared-candidate identifier;
2. a structured need descriptor;
3. a binding map;
4. a declared gap descriptor;
5. a Layout Delta descriptor over an identified Recycled Layout Asset, expressed exclusively in the operations permitted by VIII.6.8.4.

It SHALL NOT silently alter facts, provenance, or unrestricted composition structure. No output class other than these five exists at presentation time.

VIII.6.4 Instantiation and Deterministic Refinement (Normative)

Instantiation is pure binding. Refinement is limited to the governing Refinement Vocabulary, including:

- equivalence-class element substitution;
- slot rebinding and policy formatting;
- omission of declared optional elements;
- reordering explicitly permitted by the responsive grammar;
- re-execution of the pinned constraint solver.

Every refined result SHALL revalidate before display.

VIII.6.5 Session Composition Pool (Normative)

The Session Composition Pool serves Real-Time Support where needs arise unpredictably within a known topic region.

VIII.6.5.1 Pool Generation. Pool candidates MAY be generated locally during session activation and idle windows under explicit compute, memory, and candidate-count budgets. They SHALL draw only on admitted content unless the active profile and consent permit otherwise.

VIII.6.5.2 Fitness-Scored Matching. The Fitness Function SHALL produce reproducible rankings from the need descriptor, candidate metadata, pinned scoring parameters, element annotations, and locally held preference evidence. A match below threshold is a miss. The system SHALL not display a visually authoritative but semantically mediocre candidate merely because it is the best available.

VIII.6.5.3 Pool Lifecycle. Pool candidates are ephemeral. Usage and performance telemetry are Class 1 evidence. A recurring candidate pattern MAY be proposed as a persistent Recycled Layout Asset (Class 2, where derived from admitted content) or as a procedural template (Class 3) under the recurring-pattern discipline of Annex III.

VIII.6.6 Prepared-Content Miss and Staged Fallback (Normative)

Where no prepared candidate satisfies the need, the system SHALL present a correct textual core immediately and proceed according to the active profile.

A. Deterministic Profile

1. show the textual core and available admitted elements;
2. compose a validated CD from available elements, preferring a recycled layout under VIII.6.8;
3. omit or neutrally substitute missing decorative elements;
4. degrade where an informational visual cannot be represented safely;
5. queue recurring gaps for preparation.

B. Local Live Generation Profile

1. show the textual core immediately;
2. generate only declared missing elements locally;
3. reserve their layout space where possible;
4. validate each generated element;
5. reveal it atomically when complete;
6. retain its output hash and provenance.

C. External Live Generation Profile

The Local Live Generation path applies, with additional disclosure, reconstruction, and consent requirements. No external result is displayed before reconstruction and validation.

Stage-1 text is content, not a placeholder. On timeout or validation failure, the composition SHALL remain complete in a simpler Delivery Form.

VIII.6.7 Perceived Real-Time Conformance (Normative)

A presentation is perceived-real-time conformant where the operator receives a correct and useful representation without an avoidable blank waiting interval. Completion of decorative or secondary elements MAY occur within the attention window, but factual content and action boundaries SHALL already be available.

VIII.6.8 Layout Recycling and the Priority Composition Signature (Normative)

Layout Recycling is the systematic reuse of previously validated composed layouts for new content whose priority structure is the same or closely similar. It converts model-authored layout work from a per-artifact cost into a governed, accumulating library asset.

VIII.6.8.1 Priority Composition Signature. For every frame-structured composed arrangement, the system SHALL compute a Priority Composition Signature deterministically from, at minimum:

1. the ordered multiset of frame roles;
2. the normalized ranking-weight vector of the frames, quantized into declared weight bands;
3. the geometry class of each frame (area class and aspect class under the governing Frame Grammar);
4. the declared device coverage (validated size classes);
5. the governing Frame Grammar version.

The PCS computation, its quantization bands, and its similarity metric SHALL be versioned and pinned. Identical inputs SHALL yield identical signatures.

VIII.6.8.2 Storage of Recycled Layout Assets. A validated frame-structured CD MAY be generalized, during preparation or an Idle Optimization Window, into a Recycled Layout Asset: the layout structure is retained; instance content, instance provenance, and instance data bindings are stripped; typed slots remain. The asset SHALL carry its PCS, its originating-CD lineage, its validation record, its supported device matrix, and — once evaluated — its Optimization Score. Storage and staging of Recycled Layout Assets from admitted content is Class 2 with Morning Digest reporting and rollback.

VIII.6.8.3 Matching and Exact Recycling. At presentation time, the required arrangement's PCS SHALL be matched against stored Recycled Layout Assets under the pinned similarity metric.

- **Exact match** (similarity at or above the exact-match threshold): the recycled layout SHALL be instantiated by pure slot binding of the new provenance-bound content, followed by deterministic refinement under VIII.6.4 and revalidation. Exact recycling introduces no model authorship at presentation time and is fully available under the Deterministic Profile.

- **Near match** (similarity between the near-match and exact-match thresholds): the recycled layout MAY be reconciled to the required arrangement through a bounded Layout Delta under VIII.6.8.4.
- **Below near-match threshold:** the match is a miss. A structurally unsuitable layout SHALL NOT be forced onto content merely because it is the closest available.

Both thresholds SHALL be pinned, versioned configuration values.

VIII.6.8.4 The Bounded Layout Delta. A Layout Delta is authored by the Layout Model and SHALL be expressed exclusively in the following operation classes over the identified Recycled Layout Asset:

1. insertion or removal of a frame of a declared role, within grammar-permitted counts;
2. promotion or demotion of a frame between adjacent salience bands;
3. adjustment of frame geometry within the grammar's declared area and aspect classes;
4. reordering permitted by the responsive grammar;
5. slot-type-preserving slot addition or removal within a frame;
6. breakpoint-behavior adjustment within the grammar's declared responsive envelope.

A Layout Delta SHALL NOT alter facts, provenance, source association, or salience relationships mandated by VIII.8 (in particular the declared/inferred distinction), and SHALL NOT introduce structures outside the governing grammar. The delta SHALL be recorded as a structured, hash-retained object referenced by the resulting CD. The result of applying a Layout Delta SHALL pass complete revalidation — structural, responsive, accessibility, policy, and integrity — before display. A delta whose result fails validation SHALL be discarded; the system falls back to exact recycling of a different asset, staged fallback, or degradation. Layout Delta authoring under an active session is Class 2 and requires no per-instance consent dialog; it is bounded model output in the sense of VIII.6.3, not live asset generation.

VIII.6.8.5 Real-Time Correction Scope. The Layout Delta is the only mechanism by which the Layout Model modifies structure at presentation time. It corrects small deviations between the required priority structure and a recycled structure; it is not a general-purpose layout authoring path. Full layout authoring occurs in preparation, in the Session Composition Pool, or in the staged fallback.

VIII.6.8.6 Recycling Telemetry and Promotion. Recycling usage — exact-match rate, near-match rate, delta size distribution, revalidation failure rate, and per-asset reuse counts — is Class 1 evidence. Assets with sustained reuse and high Optimization Scores MAY be prioritized in matching; assets with recurrent revalidation failures SHALL be re-evaluated or deactivated under VIII.5.3.

VIII.7 Delivery Forms, Responsive Export, and Presentation Integrity

VIII.7.1 One Canonical Composition, Multiple Derivations (Normative)

The CD, its grammar, its policy set, and its hash-bound assets constitute the canonical composition. Delivery Forms are derivations of that composition; they are not independently authored artifacts.

A conforming system SHALL avoid separate semantic generation for HTML and image outputs. The same facts, textual content, element identities, and layout intent SHALL underlie every Delivery Form. Differences permitted by a Render Profile are responsive adaptation, pagination, scaling, compression, and declared omission — not semantic divergence.

Every derivation SHALL carry:

- the canonical CD hash;
- the asset-set root hash;
- CE or Rasterizer identity;
- Render Profile identifier;
- output hash;
- generation-profile provenance;
- creation time and locale;
- validation result.

VIII.7.2 Presentation Integrity Across All Forms (Normative)

The following rules apply to every Delivery Form:

1. **Validation before display.** No region is displayed before grammar, policy, provenance, and applicable accessibility checks pass.
2. **No silent factual revision.** Later correction appears as an explicit update, never as an unmarked replacement.
3. **Atomic element reveal.** A late element appears complete after validation; partially decoded or unvalidated imagery is not displayed.
4. **Reserved geometry.** Where late completion is permitted, the layout SHOULD reserve space to prevent disruptive layout shift.
5. **Source continuity.** A value and its source association SHALL remain intact across responsive transformations.
6. **Profile visibility.** Live-Derived content SHALL remain inspectably distinguishable in provenance even where it is visually integrated.
7. **Graceful degradation.** The chain is full responsive composition → reduced composition → table → text. A correct plain answer is complete; an indefinite progress indicator is not.

VIII.7.3 Responsive HTML Package — Primary Delivery Form (Normative)

Responsive HTML SHALL be the primary structured Delivery Form unless the Delivery Optimizer determines that the target cannot safely or efficiently consume it.

The package SHALL contain:

- passive semantic HTML;
- CSS within a pinned deterministic subset;
- a content-hash asset manifest;
- accessibility metadata;
- responsive rules and tested size classes;
- integrity metadata;
- no scripts, event handlers, forms, executable objects, trackers, or self-modifying content;
- no unapproved external resource references.

For the internal WR Presentation Surface, rendering SHALL cause zero network activity. Assets resolve exclusively from the local hash-addressed store.

For explicit website export, the package MAY be emitted as:

1. a self-contained bundle;
2. a content-addressed asset directory plus HTML/CSS;
3. an embeddable passive component package;
4. a server-renderable template representation;
5. a WR Anchor integration payload under VIII.7.13.

The export manifest SHALL distinguish internal WR addresses from deployable web paths. Export SHALL not leak local paths, operator identifiers, graph identifiers, private source references, or internal provenance records not intended for publication.

VIII.7.4 Responsive Behavior and Device Classes (Normative)

Every Responsive HTML Package SHALL be validated against a declared device matrix. The minimum matrix SHOULD include narrow mobile, wide mobile, tablet, desktop, and large-display classes. Frame-structured packages SHALL be validated per frame in addition to the complete edition.

Validation SHALL cover:

- content overflow;
- text expansion and localization;
- reading order;
- contrast and font scaling;
- image resolution and cropping;
- source-line retention;

- chart legibility;
- touch-target spacing where surrounding interaction is present;
- layout stability;
- print or snapshot compatibility where declared.

A responsive version is not merely a scaled desktop image. Elements MAY stack, reflow, resize, or move within grammar-declared relationships.

VIII.7.5 Raster Snapshots — Secondary Delivery Form (Normative)

A Raster Snapshot SHALL be derived by a governed Rasterizer from the canonical responsive composition — or from an individual frame — under a declared Render Profile.

The Rasterizer SHALL be tool-governed with pinned identity and content hash. The Render Profile SHALL specify at least:

- CSS viewport dimensions;
- output pixel dimensions;
- device pixel ratio;
- locale and directionality;
- font set and font hashes;
- color space;
- background and transparency;
- animation state, which SHALL be static;
- image-decoding policy;
- output format and encoder identity;
- compression and quality parameters.

The raster derivation SHALL occur only after the corresponding HTML composition validates.

VIII.7.6 Default and Optional Raster Formats (Normative)

The default raster snapshot format SHALL be WebP under a pinned encoder profile. Individual frames SHALL be exportable as WebP in every Render Profile format declared for them.

The default profile SHOULD use lossless or near-lossless settings for text-, icon-, and diagram-heavy artifacts. A domain policy MAY specify an alternative WebP profile for photo-dominant compositions.

A conforming implementation SHALL also support PNG as a lossless compatibility and archival export where required.

Implementations MAY support:

- AVIF for transfer-optimized web delivery;
- JPEG for opaque photo-dominant output;

- PDF for print and document workflows;
- SVG only where the complete composition is representable in the admitted safe vector subset.

Optional formats SHALL never replace the canonical CD and Responsive HTML Package in the audit chain.

VIII.7.7 Standard Render Profiles (Normative)

A deployment SHALL define versioned standard Render Profiles. The initial recommended set is:

- **mobile-narrow** — 360 CSS px reference width;
- **mobile-standard** — 390 CSS px reference width;
- **tablet-portrait** — 768 CSS px reference width;
- **desktop-standard** — 1440 CSS px reference width;
- **social-square** — 1080 × 1080 output;
- **social-portrait** — 1080 × 1350 output;
- **presentation-16x9** — 1920 × 1080 output;
- **print-a4** — print-profile rendering with declared margins and color policy.

These are named profiles, not hard-coded universal requirements. A deployment MAY define additional profiles, but every profile SHALL be versioned and reproducible.

VIII.7.8 Incremental Revision (Normative)

A revision SHALL operate at the smallest valid unit:

- text-slot update;
- data rebinding;
- element substitution;
- style-token update;
- frame-level recomposition;
- section-level recomposition;
- complete recomposition only where dependencies require it.

The system SHALL compute a dependency impact set and SHALL not regenerate or retransmit unaffected assets merely because one region changed.

A changed CD SHALL receive a new hash. Unchanged element hashes remain stable, permitting cache reuse and precise visual diffs.

VIII.7.9 Cross-Artifact Caching and Transfer (Normative)

Hash-addressed reusable elements SHOULD be delivered with immutable caching semantics appropriate to the deployment environment. A consuming client may retrieve an element once and reuse it across every composition referencing the same hash. Recycled

Layout Assets extend this reuse to complete layouts: repeated editions sharing a PCS reuse structure, styles, and unchanged assets, transferring only new content.

The export package SHOULD support:

- deduplicated asset manifests;
- responsive source selection;
- compression appropriate to each media class;
- lazy loading for non-critical secondary assets where consistent with presentation integrity;
- explicit preloading of first-display assets;
- cache-independent fallback behavior.

Package optimization SHALL account for shared-cache state. The byte size of one HTML file alone is not a complete lifecycle measure.

VIII.7.10 Delivery Optimizer (Normative)

The Delivery Optimizer SHALL select a Delivery Form under a deterministic policy considering:

- target capability;
- responsiveness requirement;
- accessibility requirement;
- expected reuse and cache state;
- package and transfer size;
- client rendering budget;
- offline requirement;
- archival requirement;
- privacy and network policy;
- fidelity requirement;
- integration context, including WR Anchor constraints.

Responsive HTML is preferred where reflow, semantics, accessibility, incremental revision, or reuse is valuable. A Raster Snapshot MAY be preferred where the target is static, constrained, archival, external, or where measured DOM and rendering cost exceeds the raster alternative.

The architecture SHALL not assume that HTML always consumes less client compute than a raster image. Efficiency is measured for the target process, not inferred from format labels.

VIII.7.11 Lifecycle Efficiency Measurement (Normative)

Claims of efficiency SHALL be supported by a Lifecycle Efficiency Record comparing the composed path with a declared baseline, normally an equivalent monolithic raster workflow.

Measurements SHOULD include:

- semantic-reasoning compute;
- visual-generation compute;
- time to first correct display;
- time to complete composition;
- bytes generated and stored;
- download and upload transfer volume;
- web-delivery transfer volume with cold and warm caches;
- client decode, layout, paint, and memory costs;
- revision cost for representative edits;
- cache reuse rate;
- recycling hit rate (exact and near-match) and Layout Delta cost versus full layout authoring;
- WR Anchor integration latency and update-propagation cost;
- responsive coverage;
- accessibility conformance;
- factual and visual quality scores.

An example such as a reduction from 1.5 MB to a few tens or hundreds of kilobytes MAY be documented informatively where measured. No universal factor-of-ten or energy-saving claim SHALL be presented as guaranteed without evidence for the declared workload.

VIII.7.12 Web Content Builder Use (Normative)

The Responsive HTML Package MAY be exported as a website content module. Such export SHALL preserve:

- semantic structure;
- responsive behavior;
- accessibility;
- source and attribution requirements;
- licensing constraints;
- integrity metadata;
- passive execution properties.

The export process SHALL generate a deployment report identifying:

- required assets;
- cache recommendations;
- CSS isolation strategy;
- supported browsers or rendering engines;
- any host-theme integration points;
- any features degraded for external compatibility.

This function converts the infographic generator into a structured visual content builder without changing the canonical composition model.

VIII.7.13 WR Anchor Export and Integration (Normative)

This section specifies the contract-bound integration of composed content — complete editions or individual frames — into third-party WR-connected pages. The transfer object is the Integration Contract, not a file: content, assets, and terms are agreed and committed together through the public handshake.

VIII.7.13.1 The WR Anchor and Anchor Declaration. A page participating in the WR system MAY declare one or more WR Anchors. An Anchor Declaration SHALL state:

1. a stable anchor identifier, unique within the declaring site;
2. the accepted content classes (for example: full edition, single frame, frame roles accepted);
3. the anchor's geometry class and responsive envelope;
4. the CSS isolation requirements at the anchor position;
5. the location and serving path of the Anchor Asset Directory;
6. the update policies the declaring operator is willing to agree to;
7. licensing and attribution requirements of the declaring site, where any.

The Anchor Declaration is exchanged through the initiation layer (Annex IX) or the handshake capability declaration (Annex VII). An Anchor Declaration is an invitation and identifier, not authority: nothing is transferred, hosted, or displayed on its basis alone.

VIII.7.13.2 The Integration Contract. Integration of composed content at a WR Anchor SHALL be governed by an Integration Contract concluded through the public handshake. The contract SHALL bind:

1. the exporting party and the receiving party;
2. the anchor identifier;
3. the identity (CD hash) of the integrated content block;
4. the Merkle manifest root committing the complete transferred asset set, in which every transferred file — each HTML fragment, each CSS file, and each image asset — is committed as an individual leaf, so that the presence, integrity, and completeness of every asset is cryptographically provable against the root;
5. the agreed update policy (VIII.7.13.4);
6. licensing, attribution, and export-rights terms;
7. revocation and removal terms;
8. the declared capability scope of any embedded WR Links.

The Merkle manifest model of Annex VII applies unchanged. Before conclusion, the receiving operator SHALL be shown the declared scope of the contract — content identity, asset set, update policy, embedded WR Links, and terms. Conclusion and integration then proceed as a single explicit connect-and-integrate action by the receiving operator: one recorded consent integrates the committed content block at the previously defined anchor position. Viewing an offer does not integrate it.

VIII.7.13.3 Local Asset Hosting on the Receiving Server. The transferred asset files SHALL be hosted locally on the receiving server:

1. an Anchor Asset Directory SHALL exist (or be created) on the receiving server at the path declared in the Anchor Declaration;
2. the complete asset set of the integrated block — images, CSS, and HTML fragments — SHALL be placed into that directory and served under the receiving site's own identity and origin;
3. the integrated block SHALL reference assets exclusively from the Anchor Asset Directory; it SHALL NOT reference runtime resources on the exporting deployment or any third origin (no hotlinking);
4. every hosted asset SHALL remain verifiable against its leaf hash and the contract's Merkle root; a conforming receiving implementation SHOULD verify the asset set on placement and MAY re-verify periodically;
5. the integrated block retains the passivity requirements of VIII.7.3: no scripts, no active content, no trackers, no self-modifying behavior.

In the baseline architecture, creation and population of the Anchor Asset Directory MAY be performed manually by the receiving operator following the deployment report of VIII.7.12. A later architecture stage is expressly contemplated in which directory provisioning, asset placement, and hash verification are automated by a receiving-side WR component acting under the concluded Integration Contract; such automation is a Class 3 capability at first admission and executes thereafter strictly within the contract scope.

VIII.7.13.4 The Versioned Update Path. Integrated content is hash-pinned at integration. Changes propagate only as follows:

1. a change to the integrated content SHALL be published by the exporting party as a new contract version, carrying a new CD hash and a new Merkle root, and referencing the prior root;
2. the receiving side SHALL be informed through a Content Update Notice identifying the new version, a structured summary of what changed (at the granularity of VIII.7.8: text slots, data rebinding, element substitution, frame recomposition), and any change to embedded WR Links or terms;
3. the agreed update policy determines activation: the default policy is explicit per-update confirmation by the receiving operator; a standing update scope MAY be agreed in the contract for declared change classes (for example, data rebinding within existing frames), and SHALL remain discoverable and revocable;
4. activation replaces the hosted asset set atomically with the new committed set; prior versions remain resolvable by their retained roots for audit;
5. silent replacement SHALL NOT occur: no mechanism exists by which the exporting party alters content already hosted at the anchor without a new contract version and the policy-governed activation step;

6. factual corrections SHALL be marked as explicit updates consistent with VIII.7.2 rule 2.

VIII.7.13.5 Revocation and Removal. Either party MAY terminate the Integration Contract under its terms. On termination, the receiving side removes the block and MAY remove or retain the hosted assets per the contract; contract records, roots, and Consent Records are retained under VIII.10.5. Termination of one contract does not affect other anchors or contracts.

VIII.7.13.6 Privacy, Provenance, and Interaction Boundary. The exported payload SHALL comply with the leakage prohibitions of VIII.7.3. Per-element provenance intended for publication travels with the block; internal provenance records not intended for publication do not. Embedded WR Links within an integrated block remain identifier and invitation, not authority: visitor interaction proceeds exclusively through the governed handshake of the initiation layer, and execution occurs inside the WR system, never inside the hosted block. The exporting party SHALL NOT learn how visitors of the receiving site interact with the integrated block; the receiving site's ordinary serving of its own hosted assets creates no channel back to the exporter.

VIII.8 Frame-Prioritized Editions and the Prepared Edition

VIII.8.1 Position (Normative)

This section governs every frame-structured edition: the Prepared Edition — the visual presentation form of the Morning Digest and Preparation Briefs — and real-time conversational editions of the TimesDesk™ class. It introduces no new governance object. Every Prepared Edition item remains a Digest or Brief item with evidence, provenance, mutation class, and consent affordances.

Frame-structured presentation changes presentation, not authority. Editions are assembled from validated CDs and SHOULD exhibit no operator-perceptible composition latency.

VIII.8.2 The Bayesian Priority Engine and Priority-to-Salience Mapping (Normative)

Frame arrangement SHALL derive from a local priority ranking through a deterministic mapping from ranking weight to layout weight, resolved against the governing Frame Grammar. Reasoning determines what matters and what belongs together; the grammar determines where and at what salience it appears.

The priority ranking SHALL be computed by the Bayesian Priority Engine as follows:

1. **Priors** derive from declared editorial and domain policy, preparation lineage, and — where applicable — the trigger-provenance class of the item.
2. **Evidence** comprises Priority Evidence records: declared preparation relevance; structured relevance assessments contributed by a local or cloud reasoning model as evidence inputs; locally recorded engagement telemetry; and explicit operator feedback under VIII.8.8. Model-contributed assessments enter as structured,

weighted evidence — a model SHALL NOT directly overwrite or bypass the posterior.

3. **Posterior computation** SHALL be reproducible: given the pinned prior parameters, the pinned engine version, and the recorded evidence set, the same posterior ranking results. The evidence set used for a displayed edition SHALL be identified in its audit record.
4. **Priority-to-salience mapping** is a deterministic, versioned function from posterior rank and weight to frame role, position class, area class, and typographic weight class under the Frame Grammar.

Cloud model contributions to relevance assessment are subject to the Cloud Abstraction Boundary: they operate on public, abstracted, or taxonomy-level information only. The posterior computation itself, the evidence store, and the mapping SHALL be local.

VIII.8.3 Trigger Provenance and Visual Authority (Normative)

Declared and inferred content SHALL remain visually distinguishable.

- Declared content may occupy lead positions and the full salience range.
- Inferred content SHALL appear in a distinct advisory section with a salience ceiling below the declared range.

Confidence may order inferred items but SHALL not elevate them into declared visual authority. Viewing or expanding an inferred item does not accept it. A Layout Delta SHALL NOT lift this ceiling.

VIII.8.4 Multi-Publisher Pages (Normative)

Where multiple publishers appear on one edition page:

1. ranking and salience are determined locally;
2. the host theme governs page-level coherence;
3. publisher styling is confined to the interior of its own elements;
4. per-element provenance remains discoverable;
5. no PCL field, and no publisher-supplied evidence, may influence relative prominence across publishers.

VIII.8.5 Deep-Dive Descent (Normative)

Every edition item SHALL support inspection-depth expansion under Annex VI: full content, evidence, provenance, dependencies, and prepared supporting material without new retrieval.

Focused follow-up preparation remains a separately scoped Preparation Run. Deepening a proposal does not accept it.

VIII.8.6 Assembly Budget (Normative)

Inferred edition content is bounded by the Digest budget and favors precision over recall. Declared preparation is owed completeness within its consented scope.

Full-fidelity composition is primarily warranted for declared content. Inferred content MAY be deliberately simpler, consistent with its advisory status and budget.

VIII.8.7 Delivery Forms for Editions (Normative)

Every frame-structured edition SHALL be available as responsive HTML within WR Desk™. It MAY additionally be exported — as a complete edition or as individual frames — under declared Render Profiles as WebP, PNG, PDF, or other permitted formats, and integrated at WR Anchors under VIII.7.13.

Every export remains hash-bound to the same edition CDs and SHALL preserve the declared/inferred distinction.

VIII.8.8 Feedback Evidence and Ranking Refinement (Normative)

The system SHALL provide explicit feedback affordances on frames and composed graphics, including at minimum an approval signal, a disapproval signal, and optional free-text description.

1. **Recording.** Each feedback event SHALL be recorded locally as a structured Priority Evidence record: subject (frame or graphic identity and CD hash), signal, timestamp, and — for free text — a locally parsed structured representation alongside the retained original text. Feedback recording is Class 1.
2. **Effect.** Feedback evidence enters the Bayesian Priority Engine (VIII.8.2), the Fitness Function (VIII.3.8), and the Optimization Score evaluation (VIII.11) through their pinned evidence channels. An evidence-driven change to ranking parameters or scoring weights is Class 2 and reportable.
3. **Refinement loops.** Automated research-inspired loops MAY refine the Bayesian Priority Engine's parameters — priors, evidence weights, quantization bands — exclusively in Idle Optimization Windows under VIII.11.2, with fixed evaluation sets, declared hypotheses, measured results, and rollback. The presentation path never trains, tunes, or refits ranking parameters.
4. **Privacy.** Feedback evidence is local evidence. It SHALL NOT be observable by publishers and SHALL NOT cross the Cloud Abstraction Boundary except in abstracted, taxonomy-level form under an explicitly authorized disclosure path.

VIII.9 The Augmented Overlay

VIII.9.1 Position and Toggle Discipline (Normative)

The Augmented Overlay is an interactive layer over passive composed artifacts. It provides annotations, navigation, automation suggestions, and spoken or pointer-based affordances.

It is operator-toggleable globally, per Presentation Surface, and per context region. No publisher, PCL, grant, or delivered artifact may activate it, prevent deactivation, or condition core functionality on its use.

VIII.9.2 Overlay Content (Normative)

Overlay content is composed and provenance-bound. Its passive visual layer is represented by CDs and rendered under the same HTML/CSS restrictions as ordinary artifacts. Interaction logic belongs to the trusted host runtime, not the exported artifact.

Publisher overlay content is Class 3 and region-bound at admission. Derived overlay annotations follow the Derived asset rules. New automation paths remain governed procedural capabilities.

VIII.9.3 Automation Suggestions and Execution Boundary (Normative)

An overlay suggestion pre-satisfies nothing. Every consequential action remains gated by PoAE™ at the point of effect.

Prepared actions retain their pinned template, schema validation, precise effect display, consent tap, and PoAE™ record. Overlay placement does not alter their authority.

VIII.9.4 Interaction Modalities and Evidence (Normative)

Speech SHALL be transcribed locally. Spoken confirmation is input, not an authorization factor unless a separate PoAE™ rule explicitly defines it.

Pointer, dismissal, expansion, and engagement telemetry are local evidence and SHALL not be observable by publishers. Ignored or dismissed content MAY dampen future surfacing.

The overlay MAY provide “prepare this” and “deepen this” affordances. These create evidence, declarations, or proposals through ordinary governance; they do not bypass preparation or training boundaries.

VIII.9.5 Pre-Export Editable Regions (Normative)

A composed artifact intended for export MAY declare Editable Regions: specially marked slots, elements, or frames that SHALL be modifiable through Augmented Overlay interaction before export. Where Editable Regions are declared, a conforming implementation SHALL support both modification modes:

- **Manual editing.** The operator edits the marked region directly through the overlay: text-slot content, bound values within domain-policy constraints, element substitution within declared equivalence classes, or removal of declared optional elements.
- **AI-assisted editing.** The operator requests a model-proposed revision of the marked region. The proposal is authored by the local models of this Annex (the Layout Model for structural aspects; grammar-constrained textual generation for text), is confined to the marked region, and is presented for explicit operator acceptance before it takes effect. A proposal never applies itself.

The following rules apply to every edit, in both modes:

- the edit operates at the smallest valid revision unit of VIII.7.8 and produces a revised CD with a new hash; unchanged element hashes remain stable;
- factual values continue to bind from provenance-bound sources; an edit that would detach a value from its source association SHALL be rejected or SHALL downgrade the value's presentation to unbound operator content with distinct provenance (`local:` binding);
- structural modifications remain within the governing grammar; AI-assisted structural edits are bounded model output in the sense of VIII.6.3;
- the revised CD SHALL pass complete revalidation before display and before any export; export of an artifact with pending unvalidated edits SHALL NOT occur;
- every edit is recorded — region, mode (manual or AI-assisted), before/after hashes, and, for AI-assisted edits, the proposal record and the operator's acceptance;
- Editable Region markings are composition metadata: they SHALL NOT appear as interactive affordances in any exported Delivery Form, which remains passive under VIII.7.3.

Editable Regions change what the operator can conveniently revise before publication; they change no authority. Overlay editing of a region is preparation and composition, never execution.

VIII.9.6 Overlay Export (Normative)

Website, WR Anchor, and raster exports SHALL exclude runtime overlay behavior unless the operator explicitly requests a static annotation layer. Static annotation export SHALL be flattened into the CD and lose all execution affordances.

VIII.10 Governance, Privacy, Auditability, and PoAC™ Classification

VIII.10.1 Grant Model and Execution Boundary (Normative)

All activity under this Annex is preparation, composition, delivery, or presentation. PCL delivery and updates ride on delivery/preparation rights. No CD, asset, Delivery Form, salience level, recommendation, Integration Contract, or overlay item grants execution authority.

A prepared action remains executable only through the applicable human authorization and PoAE™ record.

VIII.10.2 Privacy and External Interaction (Normative)

All external interaction in service of composition is subject to Annex III §7 and Annex VI §VI.7.

Under the Deterministic Profile, presentation-time rendering causes no external interaction.

Under External Live Generation, the specific disclosure SHALL be authorized before transmission. The system SHALL minimize disclosure, prefer taxonomy-level or masked requests, and perform mandatory local instantiation where the returned result is generic.

A publisher SHALL not learn which library elements are used, when a composition is displayed, which Render Profile is selected, which layouts are recycled, or how the operator interacts, except where the operator independently publishes or transmits an exported artifact or concludes an Integration Contract, in which case only the contract-scoped information is shared.

VIII.10.3 Auditability and Reproduction (Normative)

For every displayed or exported artifact, the system SHALL retain or be able to resolve:

- the canonical CD;
- grammar and domain-policy versions;
- asset-set hashes;
- the recycled-layout reference, its PCS, and the Layout Delta hash where recycling occurred;
- the Priority Evidence set identity and Bayesian engine version where the artifact is a ranked edition;
- CE and Rasterizer identities;
- Render Profile;
- generation profile and authorization reference;
- preparation lineage;
- validation results;
- output hash;
- the Integration Contract identity and Merkle root for anchor-integrated content;
- explicit updates or invalidations.

Under the Deterministic Profile, the artifact SHALL be exactly reproducible from retained inputs; where a Layout Delta was applied, reproduction replays the retained delta object, not the model.

Under a live-generation profile, the exact generative process MAY be non-reproducible. In that case, the generated output actually displayed SHALL be retained by hash with sufficient provenance to reproduce the final artifact, and the audit record SHALL state that process-level regeneration is not guaranteed.

VIII.10.4 PoAC™ Assignment Table (Normative)

Operation	Class
CD rendering from validated inputs	1
Raster derivation under a pinned Render Profile	1
Responsive validation and static	1

Operation	Class
conformance tests	
Session Pool generation, scoring, refinement, and discard from admitted content	1
PCS computation and recycled-layout matching	1
Presentation and lifecycle telemetry recording, including recycling and feedback telemetry	1
Feedback evidence recording	1
Evidence-based element re-ranking	2
Evidence-driven update of Bayesian Priority Engine parameters	2
Candidate Set and edition assembly from admitted content	2
Local Derived asset generation and staging from admitted content, including Recycled Layout Assets	2
Layout Delta authoring and application under an active session	2
Idle-window optimization of stored layouts with Optimization Score assignment	2
Manual edit of a declared Editable Region before export	1
AI-assisted edit of a declared Editable Region (proposal + accepted application)	2
Deactivation and dependent cascade of a Derived asset	2
Local Live-Derived generation under an existing scoped authorization	2
Persistent promotion of a local Live-Derived asset from admitted content	2
Content update publication and activation under a standing update scope of an Integration Contract	2
PCL acquisition and admission	3
External Live Generation disclosure and interaction	3, unless covered by an explicit standing scope valid for the request
Persistent admission of externally	3

Operation	Class
generated content	
New Source ingestion	3
Conclusion of an Integration Contract (export or receipt)	3
First admission of automated Anchor Asset Directory provisioning	3
Promotion of a recurring pool pattern to a procedural template	3
Domain-policy override	3
Admission of publisher overlay content or new automation templates	3

Nothing in this table permits new external knowledge or capability to enter the WR Graph™ without the Consent Record required by Annex III.

VIII.10.5 Retention and Deletion (Normative)

Audit retention SHALL be proportional to assurance requirements and applicable law. Deactivation SHALL be preferred to destructive deletion where prior artifacts depend on a retained hash. Integration Contract records and Merkle roots SHALL be retained for the life of the contract and thereafter per retention policy.

Where an operator requests deletion of Live-Derived content, the system SHALL explain any conflict with legal, security, or audit retention and SHALL remove future-use references immediately where permitted.

VIII.11 Quality-Constrained Optimization and Evaluation

VIII.11.1 Optimization Objective (Normative)

Optimization SHALL seek better allocation of resources at equal or higher output quality. The optimization objective is multi-dimensional and SHALL not collapse into minimum file size or minimum latency alone.

The objective vector SHOULD include:

283. factual correctness;
284. semantic coverage;
285. visual hierarchy;
286. legibility;
287. accessibility;
288. responsive stability;
289. operator comprehension;
290. provenance completeness;

291. generation latency;
292. inference and rendering compute;
293. package and transfer size;
294. cache and recycling reuse;
295. revision cost;
296. integration effort.

Hard Quality Floor constraints dominate soft efficiency objectives.

VIII.11.2 Idle-Window Optimization Discipline and Research-Inspired Loops (Normative)

A deployment MAY run automated experimentation loops over admissible variants of:

- retrieval parameters;
- element ranking;
- Bayesian Priority Engine parameters;
- recycled-layout selection and PCS quantization bands;
- layout selection;
- responsive constraints;
- load-time characteristics of stored layouts (asset ordering, preload sets, compression, structural simplification within the grammar);
- compression profiles;
- Render Profile parameters;
- cache and preload policy;
- layout grammar refinements.

Every experiment SHALL have:

- a declared hypothesis;
- a fixed evaluation set;
- quality and safety constraints;
- reproducible parameters;
- measured results;
- rollback;
- governance classification;
- a prohibition on autonomous expansion into new external Sources or capabilities.

Optimization loops execute exclusively in Idle Optimization Windows. The presentation path SHALL NOT execute optimization experiments, parameter fitting, or training of any kind. An optimized layout variant SHALL be stored back into the Element Library as a versioned Recycled Layout Asset (or a new version of one), indexed by its PCS and carrying its measured Optimization Score; the presentation path consumes stored, validated results only.

The loop may discover better parameters. It does not grant itself new authority.

VIII.11.3 Accessibility and Localization (Normative)

Responsive HTML output SHALL support semantic reading order, selectable text, alternate descriptions, sufficient contrast, and keyboard-compatible surrounding controls.

Localization tests SHALL account for text expansion, right-to-left direction, locale-specific number and date formats, and font coverage. A layout valid only for the language in which it was generated is not generally responsive.

Raster exports SHOULD embed accessible descriptions in surrounding document metadata where the target format permits. The existence of a raster export does not waive the requirement to preserve an accessible structured form where available.

VIII.11.4 Security Evaluation (Normative)

Export and rendering tests SHALL include:

- script and active-content exclusion;
- external-reference detection, including verification that anchor-integrated blocks reference only their Anchor Asset Directory;
- CSS subset conformance;
- font and media validation;
- path and identifier leakage checks;
- reconstruction enforcement;
- content-security-policy compatibility;
- hash-manifest verification, including Integration Contract Merkle-root verification;
- malformed-layout and resource-exhaustion tests.

VIII.11.5 Benchmark Baselines (Normative)

A claimed improvement SHALL name its baseline. Suitable baselines include:

- full-image generative synthesis;
- manually authored raster infographic;
- static raster web embedding;
- separately authored desktop and mobile artifacts;
- per-artifact full layout authoring without recycling;
- file-based syndication without contract-bound integration;
- cold-cache and warm-cache delivery.

Quality equivalence SHALL be evaluated independently of efficiency. A smaller artifact that communicates less, obscures evidence, or fails accessibility is not an improvement.

VIII.11.6 Conformance Levels (Normative)

A deployment MAY declare:

- **Composition Conformance:** valid CD, Element Library, deterministic CE;

- **Responsive Delivery Conformance:** passive responsive HTML and device-matrix validation;
- **Multi-Form Conformance:** governed raster derivation and Render Profiles;
- **Edition Conformance:** frame structure, Bayesian Priority Engine, deterministic priority-to-salience mapping, and feedback evidence;
- **Recycling Conformance:** PCS computation, recycled-layout matching, bounded Layout Delta with revalidation, and idle-window optimization;
- **Anchor Integration Conformance:** Anchor Declarations, Integration Contracts with complete Merkle asset commitment, local asset hosting, and the versioned update path;
- **Lifecycle Optimization Conformance:** measurement, incremental revision, and cache-aware delivery;
- **WR Deterministic Conformance:** default profile, exact reproduction, zero render-time external interaction;
- **WR Informed Live Generation Conformance:** scoped operator choice, differentiated local/external modes, validation, provenance, and revocation.

A higher declaration SHALL include the requirements of lower levels on which it depends.

VIII.12 Relation to Prior Work, General Adoption, and Claimed Contribution (Informative)

Declarative UI systems, HTML/CSS rendering, design systems, component libraries, responsive web design, visualization grammars, content-addressed storage, image rasterization, recommender systems, Bayesian ranking, constrained decoding, caching, precomputation, and multi-channel publishing are established practice and are not individually claimed as novel. Likewise, responsive auto-layout mechanisms, placeholder-field population, WebP output, accessibility standards, and media queries are treated here as common professional groundwork, not as inventive substance.

The general architectural contribution expressed by this Annex is their coordinated application to generative visual information:

- treating the composition rather than the monolithic image as the canonical artifact, with all composition state — including content-to-position association — held inside one schema-validated Composition Description rather than in any separately stored control specification;
- assigning model reasoning to information, selection, and arrangement, with the responsive arrangement authored by a locally operated, layout-specialized model rather than chosen from a fixed catalog of pre-authored page designs;
- retrieving optimized visual elements from an annotated library;
- detecting gaps and confining generation to genuinely missing elements;
- structuring editions as prioritized frames whose salience derives deterministically from a reproducible Bayesian ranking over local evidence, including explicit operator feedback;

- recycling previously validated, model-authored layouts by Priority Composition Signature, with bounded model-authored correction and mandatory revalidation — reuse of emergent layout artifacts, not selection from authored template galleries;
- continuously improving the library through measured, quality-constrained loops confined to idle windows;
- deriving responsive HTML and multiple image or document forms from one composition, at edition and at frame granularity;
- integrating composed content into third-party pages as a contract object — every asset Merkle-committed, hosted under the receiving site's own identity, and updated only through explicit contract versions — rather than as an opaque syndicated file or a channel-presentation slot within one operator's system;
- reducing revision, integration, syndication, transfer, and reuse costs across the complete lifecycle;
- permitting delivery optimization by target rather than assuming one format is universally best.

The WR-specific contribution is the governance layer applied to that architecture:

- attestation-bound library admission;
- per-element provenance;
- reconstruction of media-class assets;
- deterministic prepared-first presentation;
- a strict but non-paternalistic default;
- separate local and external live-generation profiles;
- informed, scoped, revocable operator choice;
- explicit handling of reduced reproducibility;
- trigger-provenance binding to visual authority, unliftable by any layout operation;
- cross-publisher ranking exclusivity and publisher-blind interaction telemetry;
- consent-concluded Integration Contracts with cryptographically provable asset sets and a no-silent-replacement update path;
- exact audit of what was shown, from what, under which rules.

Large LLM and visual-generation providers may adopt the general architecture without adopting WR governance. In that environment, real-time generation of missing elements may be routine. The essential principle remains: **generate the irreducibly new part; retrieve, recycle, and compose the rest; derive every required form from one structured source; and syndicate as a verifiable contract, not an opaque file.**

VIII.13 Version History

Version	Date	Change
1.0	2026-07-11	Initial normative Hybrid Composition specification: preparation/presentation compute split; Inversion

Version	Date	Change
		Rule; Composition Description; passive HTML/CSS rendering; PCL and Derived Composition Assets; Candidate Sets; Session Composition Pool; staged fallback; Prepared Edition; Augmented Overlay; PoAC™ classification; privacy and auditability.
1.1	2026-07-14	Terminology alignment with WR Advance Preparation, WR Predictive Preparation, Real-Time Support, and Private Distribution Path. No intended semantic change.
2.0	2026-07-17	Full lifecycle revision: problem statement for monolithic raster infographics; general applicability beyond WR; annotated, optimization-driven element retrieval; gap detection and generation limited to missing elements; WR Deterministic, Local Live Generation, and External Live Generation profiles; Live-Derived provenance and admission; responsive HTML as primary Delivery Form; WebP raster snapshots with PNG compatibility and versioned Render Profiles; website content-builder export; incremental revision; cross-artifact caching; Delivery Optimizer; lifecycle-efficiency measurement; research-inspired quality-

Version	Date	Change
		constrained optimization; accessibility, localization, and security evaluation; PoAC™ classification.
3.0 (Final)	2026-07-18	Final publication revision. Adds frame structure as a first-class CD construct with per-frame responsiveness and per-frame export (HTML fragment and WebP); the Bayesian Priority Engine with reproducible posterior ranking, model-contributed evidence, operator feedback evidence (approval/disapproval and parsed free text), and idle-only refinement loops; the Priority Composition Signature and Layout Recycling with exact-match pure rebinding and the bounded, revalidated Layout Delta as the fourth permitted presentation-time exception (VIII.1.7, VIII.6.3 extended to five bounded output classes); Recycled Layout Assets as governed Derived assets with Optimization Scores; the idle-window optimization discipline for load-time and responsive optimization; WR Anchor export and integration: Anchor Declarations, Integration Contracts with complete per-asset Merkle manifest commitment, local asset hosting in the receiving site’s Anchor Asset Directory (manual

Version	Date	Change
		baseline, automatable provisioning contemplated as Class 3), the versioned no-silent-replacement update path, and revocation; pre-export Editable Regions modifiable through Augmented Overlay interaction, manually or AI-assisted, with mandatory revalidation before export (VIII.9.5); introduction extended: text-or-image status quo of chat assistants and TimesDesk™ as a hybrid local/cloud merge with massive parallelization; privacy through obfuscation, masking, and distribution obfuscation with locally operated recommender systems; handshake-contracted external automation with pipeline-enforced deterministic execution; voluntary, data-local WR Ads cost-offset cross-reference; syndication loss added to the lifecycle problem statement; second essential combined disclosure; updated PoAC™ table, audit requirements, conformance levels, benchmarks, and measurements.

3.1 — 2026-07-18 — Hybrid sharpening: the hybrid local/cloud split stated explicitly as the defining architectural principle across all facets and artifact classes; office-format artifacts (word-processing documents, spreadsheets) integrated over handshake-transferred text-pure data with mandatory local generation and interpretation (no foreign document containers cross the boundary); the response surface characterized as audiovisual (pointer/cursor interaction via the Augmented Overlay, locally transcribed speech input, video via attested/declared media references such as platform-hosted videos); new informative outlook VIII.1.10 on handshake-conveyed computer-use execution with pre-declared chains and model-as-router, expressly marked as subject to evaluation.

License Notice

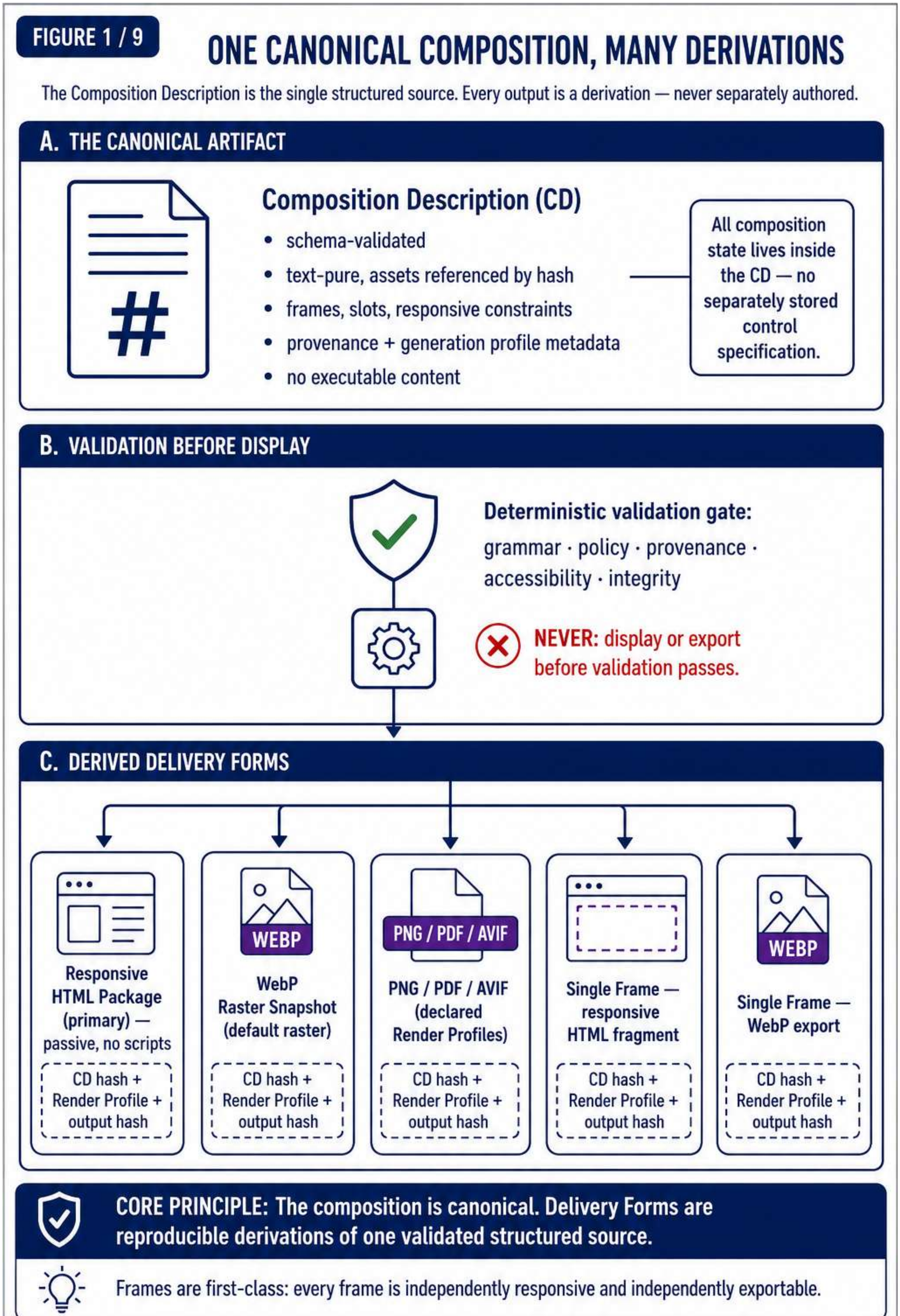


FIGURE 1 / 9
ONE CANONICAL COMPOSITION, MANY DERIVATIONS
The Composition Description is the single structured source. Every output is a derivation — never separately authored.
A. THE CANONICAL ARTIFACT
#
Composition Description (CD)
• schema-validated
• text-pure, assets referenced by hash
• frames, slots, responsive constraints
• provenance + generation profile metadata
• no executable content
All composition state lives inside the CD — no separately stored control specification.
B. VALIDATION BEFORE DISPLAY
Deterministic validation gate:
grammar · policy · provenance · accessibility · integrity
NEVER: display or export before validation passes.
C. DERIVED DELIVERY FORMS
Responsive HTML Package (primary) — passive, no scripts
CD hash + Render Profile + output hash
WEBP
WebP Raster Snapshot (default raster)
CD hash + Render Profile + output hash
PNG / PDF / AVIF
PNG / PDF / AVIF (declared Render Profiles)
CD hash + Render Profile + output hash
Single Frame — responsive HTML fragment
CD hash + Render Profile + output hash
WEBP
Single Frame — WebP export
CD hash + Render Profile + output hash
CORE PRINCIPLE: The composition is canonical. Delivery Forms are reproducible derivations of one validated structured source.
Frames are first-class: every frame is independently responsive and independently exportable.

FIGURE 2 / 9

THE FRAME-PRIORITIZED NEWSPAPER EDITION

A reproducible Bayesian ranking decides what matters; a deterministic mapping decides where and how large it appears.

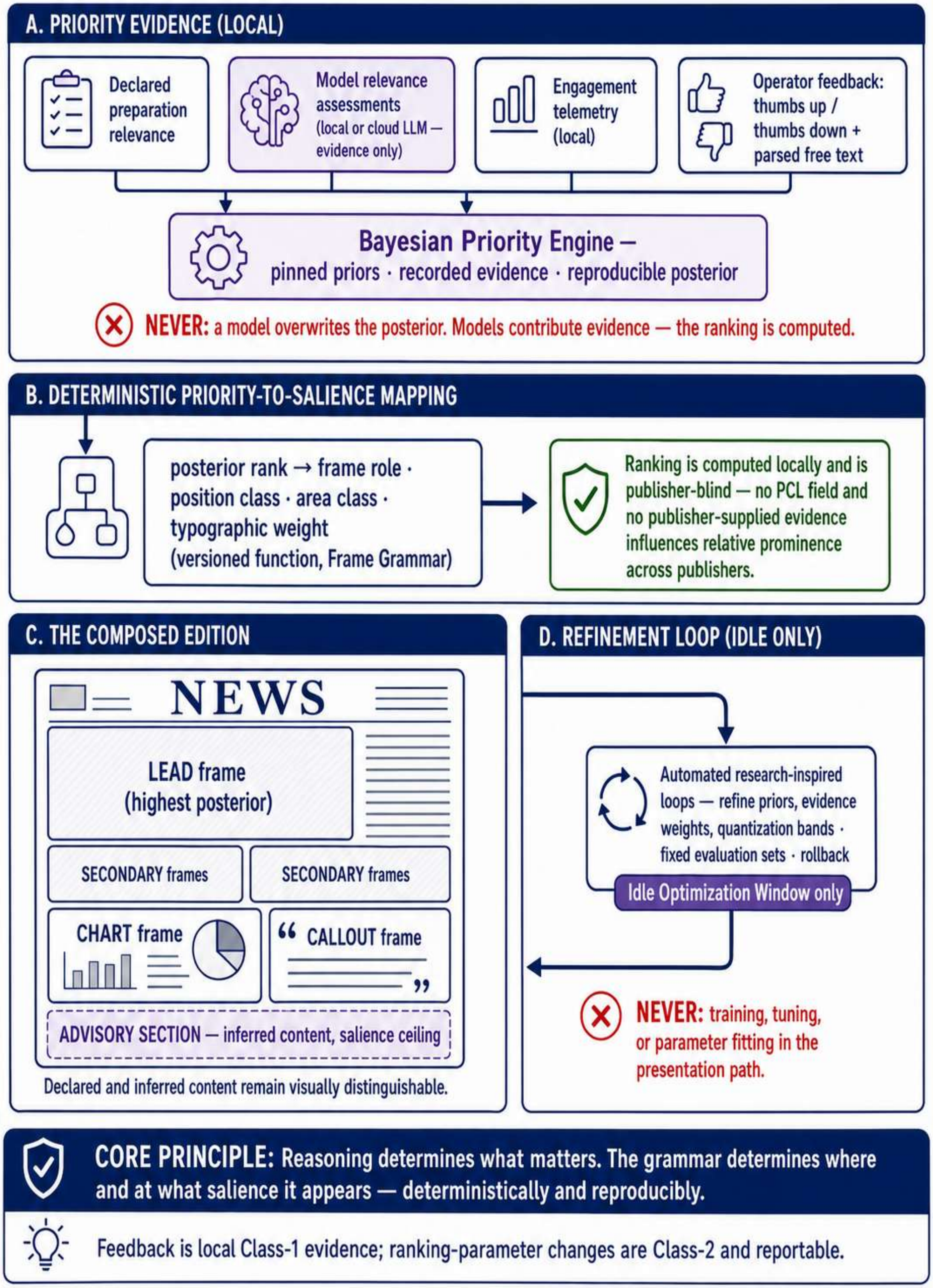


FIGURE 3 / 9

LAYOUT RECYCLING BY PRIORITY COMPOSITION SIGNATURE

Model-authored layouts become governed, reusable library assets — indexed by the priority structure they express.

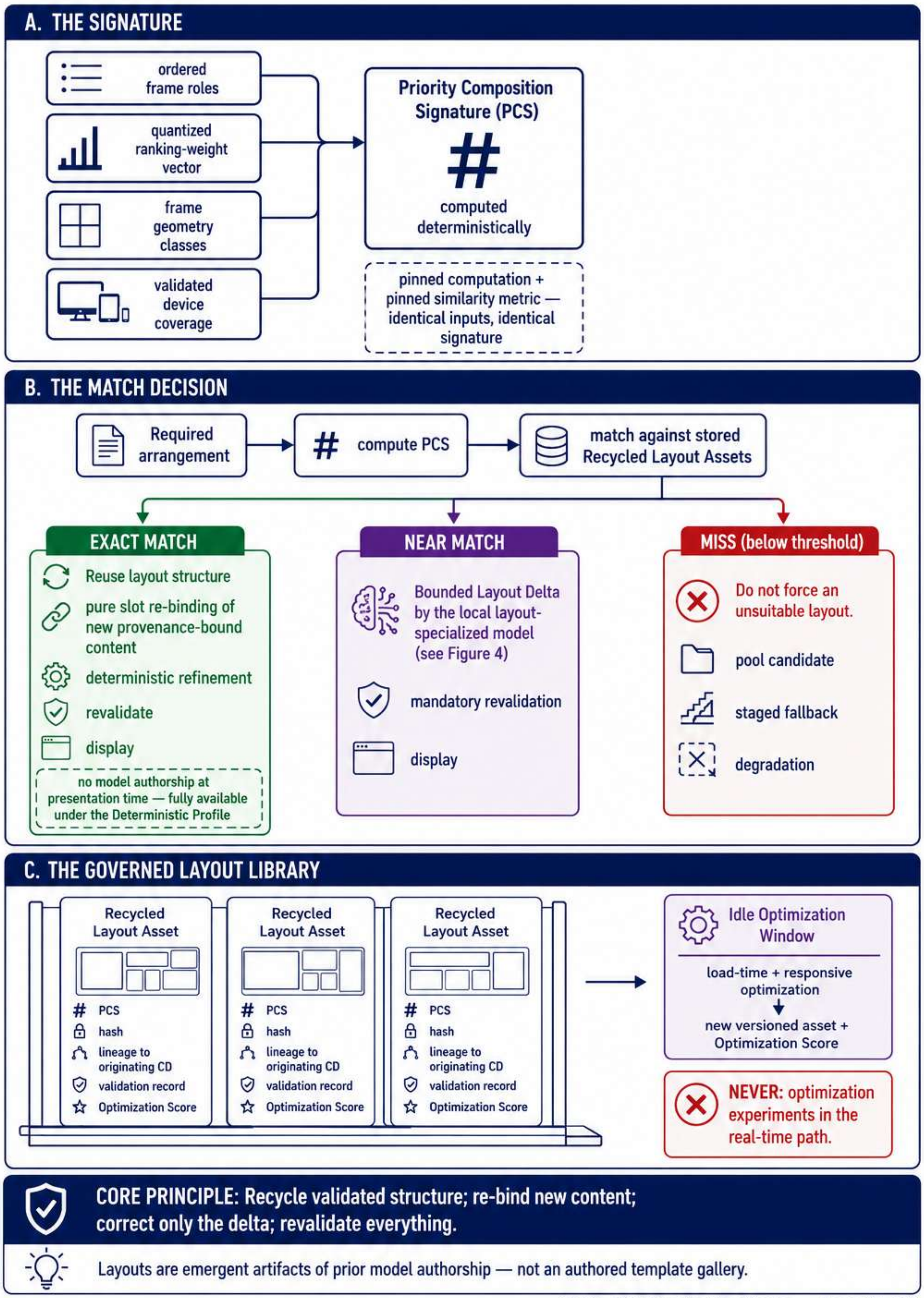


FIGURE 4 / 9

THE BOUNDED LAYOUT DELTA

The only structural model output at presentation time — closed operation set, mandatory revalidation, hash-retained.

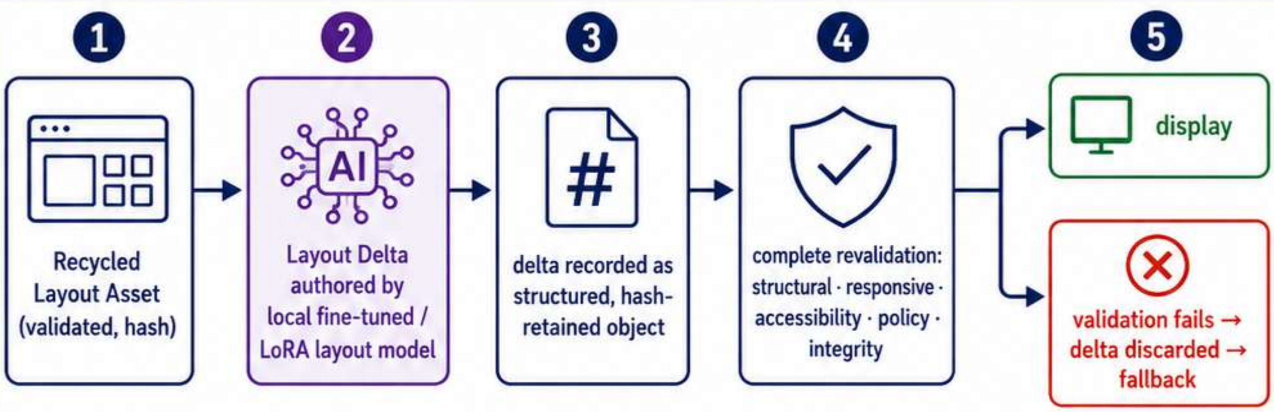
A. PERMITTED OPERATIONS (CLOSED SET)

- insert / remove a frame of a declared role (within grammar counts)
- promote / demote between adjacent salience bands
- adjust geometry within declared area + aspect classes
- reordering permitted by the responsive grammar
- slot-type-preserving slot addition / removal
- breakpoint behavior within the responsive envelope

B. PROHIBITED, ALWAYS

NEVER: alter facts · alter provenance · break source association · lift the inferred-content salience ceiling · introduce structure outside the grammar

C. THE PIPELINE



D. REPRODUCTION

Audit replay uses the retained delta object — never the model.
Deterministic Profile reproduction is exact.

CORE PRINCIPLE: Bounded selection, bounded text, staged fallback, bounded Layout Delta — the four and only four presentation-time exceptions to determinism.

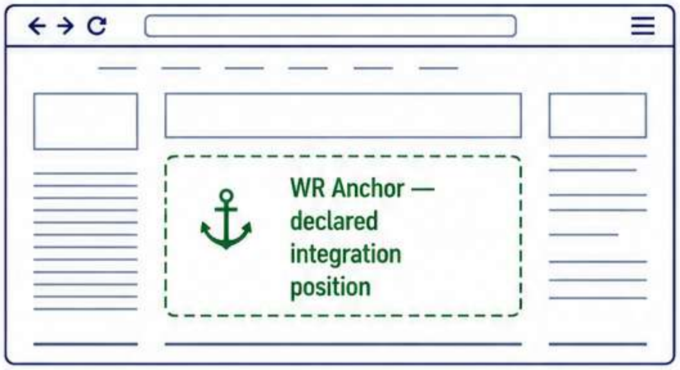
A delta is bounded model output (Class 2), not live asset generation — no per-instance consent dialog required.

FIGURE 5 / 9

WR ANCHOR INTEGRATION: THE CONTRACT IS THE TRANSFER OBJECT

Composed content enters a third-party page as a cryptographically committed contract — not as an opaque file.

A. THE ANCHOR DECLARATION



ANCHOR DECLARATION

- anchor identifier
- accepted content classes
- geometry class
- CSS isolation
- Anchor Asset Directory path
- offered update policies

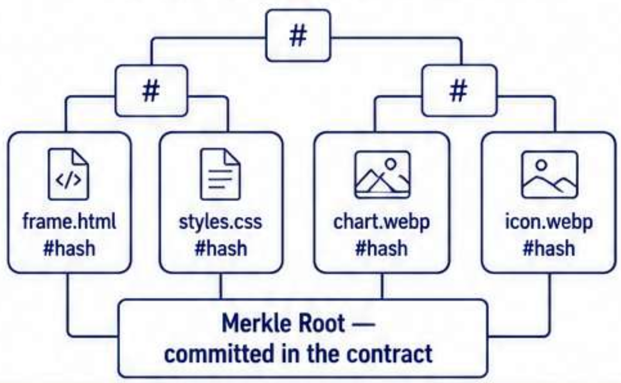
An Anchor Declaration is identifier and invitation — not authority.

B. THE INTEGRATION CONTRACT (PUBLIC HANDSHAKE)

INTEGRATION CONTRACT

- parties
- anchor id
- CD hash
- Merkle root
- update policy
- licensing + attribution
- revocation terms
- WR Link scope

MERKLE TREE OF COMPOSED ASSETS



C. CONNECT AND INTEGRATE

1. receiving operator is shown the declared scope
2. one connect-and-integrate action (recorded consent)
3. asset set placed into the Anchor Asset Directory on the receiving server
4. block served under the receiving site's own identity — verified against leaf hashes + root

D. HOSTING RULES

- assets hosted locally on the target server
- verifiable against the contract root
- passive block: no scripts, no trackers

- **NEVER:** hotlinking to the exporter
- **NEVER:** runtime dependency on the exporting deployment
- **NEVER:** visitor telemetry back to the exporter

Baseline: directory created manually.
Contemplated later stage: automated provisioning under the concluded contract (Class 3 at first admission).

CORE PRINCIPLE: The page invites, the handshake agrees, the content is committed and locally hosted — provable against one Merkle root.

Embedded WR Links remain identifier and invitation; execution happens inside the WR system, never inside the hosted block.

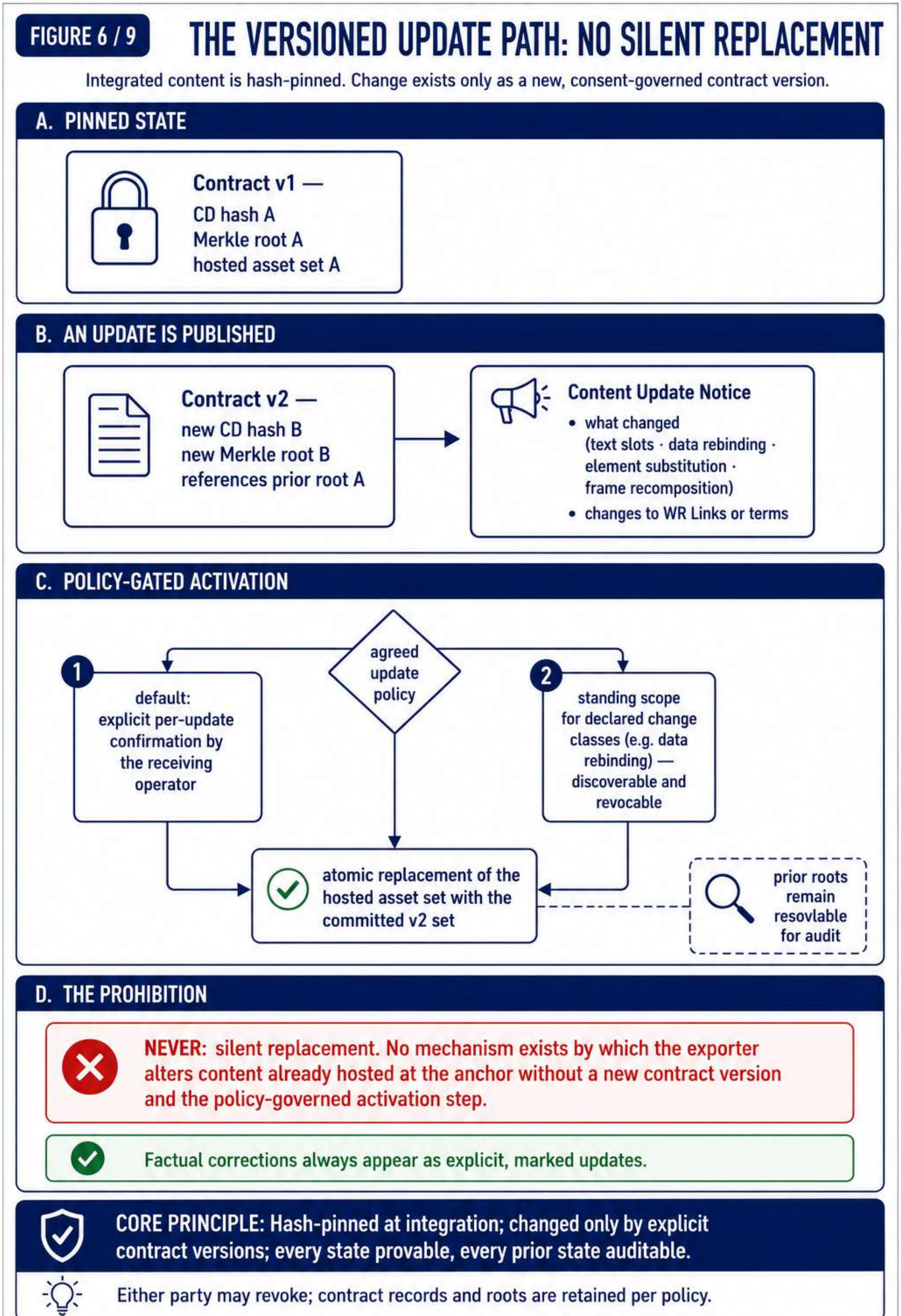


FIGURE 6 / 9
THE VERSIONED UPDATE PATH: NO SILENT REPLACEMENT
Integrated content is hash-pinned. Change exists only as a new, consent-governed contract version.
A. PINNED STATE
Contract v1 —
CD hash A
Merkle root A
hosted asset set A
B. AN UPDATE IS PUBLISHED
Contract v2 —
new CD hash B
new Merkle root B
references prior root A
Content Update Notice
• what changed (text slots · data rebinding · element substitution · frame recomposition)
• changes to WR Links or terms
C. POLICY-GATED ACTIVATION
agreed update policy
1
default: explicit per-update confirmation by the receiving operator
2
standing scope for declared change classes (e.g. data rebinding) — discoverable and revocable
atomic replacement of the hosted asset set with the committed v2 set
prior roots remain resovlable for audit
D. THE PROHIBITION
NEVER: silent replacement. No mechanism exists by which the exporter alters content already hosted at the anchor without a new contract version and the policy-governed activation step.
Factual corrections always appear as explicit, marked updates.
CORE PRINCIPLE: Hash-pinned at integration; changed only by explicit contract versions; every state provable, every prior state auditable.
Either party may revoke; contract records and roots are retained per policy.

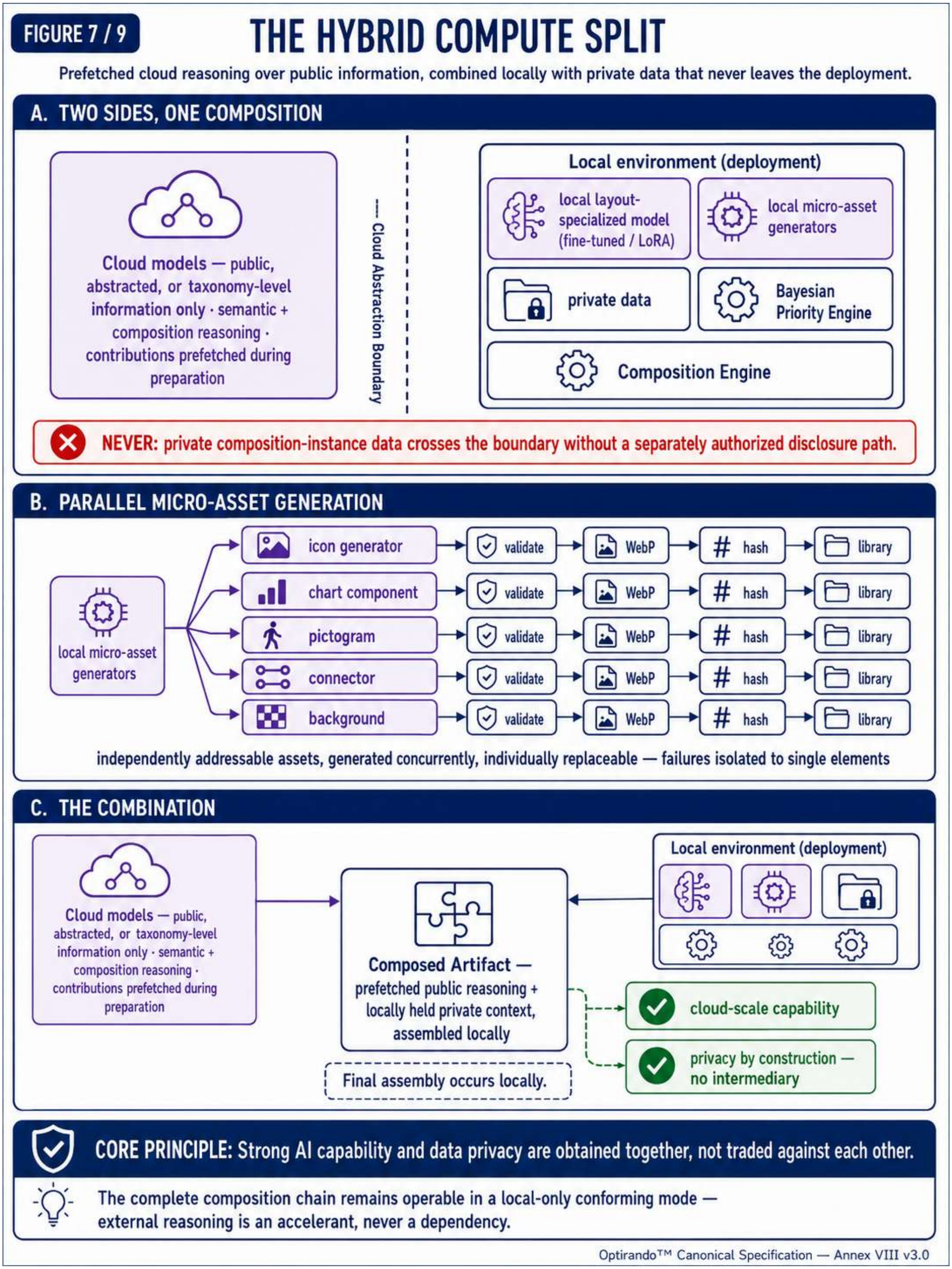


FIGURE 7 / 9
THE HYBRID COMPUTE SPLIT
Prefetched cloud reasoning over public information, combined locally with private data that never leaves the deployment.
A. TWO SIDES, ONE COMPOSITION
Cloud models — public, abstracted, or taxonomy-level information only · semantic + composition reasoning · contributions prefetched during preparation
Cloud Abstraction Boundary
Local environment (deployment)
local layout-specialized model (fine-tuned / LoRA)
local micro-asset generators
private data
Bayesian Priority Engine
Composition Engine
NEVER: private composition-instance data crosses the boundary without a separately authorized disclosure path.
B. PARALLEL MICRO-ASSET GENERATION
local micro-asset generators
icon generator
chart component
pictogram
connector
background
validate
WebP
hash
library
independently addressable assets, generated concurrently, individually replaceable — failures isolated to single elements
C. THE COMBINATION
Cloud models — public, abstracted, or taxonomy-level information only · semantic + composition reasoning · contributions prefetched during preparation
Composed Artifact — prefetched public reasoning + locally held private context, assembled locally
Local environment (deployment)
cloud-scale capability
privacy by construction — no intermediary
Final assembly occurs locally.
CORE PRINCIPLE: Strong AI capability and data privacy are obtained together, not traded against each other.
The complete composition chain remains operable in a local-only conforming mode — external reasoning is an accelerant, never a dependency.
Optirando™ Canonical Specification — Annex VIII v3.0

FIGURE 8 / 9

GOVERNANCE OF THE VISUAL CHAIN

Provenance for every element, informed choice for every generation, proof for everything displayed.

A. PER-ELEMENT PROVENANCE CLASSES

- # host:
- # pcl: (admitted publisher artifact)
- # derived: (incl. Recycled Layout Assets)
- # live-derived-local:
- # live-derived-external:

Every displayed element resolvable by hash.
No class carries elevated factual authority — facts always bind from declared sources.

B. GENERATION PROFILES (OPERATOR CHOICE)

WR Deterministic Profile
(default)

- no presentation-time synthesis
- exact reproduction
- zero render-time external interaction

Local Live Generation
(optional)

- only missing elements
- validated before display
- scoped + revocable

External Live Generation
(optional, highest disclosure)

- informed dialog: what leaves, to whom, cost, latency, reproducibility

Declining live generation never loses the textual answer.

NEVER: viewing, waiting, or continuing a conversation counts as authorization.

C. THE AUDIT RECORD

AUDIT LEDGER

- canonical CD + grammar/policy versions
- # asset-set hashes
- recycled-layout ref + PCS + Layout Delta hash
- Priority Evidence set + engine version
- CE/Rasterizer identity
- Render Profile
- Integration Contract + Merkle root
- validation results
- # output hash

CORE PRINCIPLE: The system prepares, the human authorizes, and the system records cryptographic proof.

Suggestion never pre-satisfies authorization — composed artifacts inform; they never execute.

FIGURE 9 / 9 PRE-EXPORT EDITING VIA THE AUGMENTED OVERLAY

Specially marked regions are modifiable before export — manually or AI-assisted — and nothing leaves without revalidation.

A. MARKED REGIONS IN THE COMPOSITION

Editable Region — declared in the CD

Editable Region markings are composition metadata — they never appear as interactive affordances in any exported Delivery Form; the export stays passive.

B. TWO MODIFICATION MODES

Augmented Overlay interaction

MANUAL EDITING — operator edits directly:

- text-slot content
- bound values within domain policy
- element substitution within equivalence classes
- removal of declared optional elements

AI-ASSISTED EDITING —

- model-proposed revision, confined to the marked region
- Layout Model for structure
- grammar-constrained text generation
- presented for explicit operator acceptance

NEVER: a proposal applies itself.

C. THE EDIT PIPELINE

1. edit at the smallest valid revision unit
2. revised CD, new hash — unchanged element hashes stay stable
3. facts keep their source binding (an edit that would detach a value is rejected, or the value is downgraded to local: operator content with distinct provenance)
4. complete revalidation
5. export permitted

pending unvalidated edits → export blocked

NEVER: export with pending unvalidated edits.

D. THE EDIT RECORD

EDIT LEDGER

- region
- mode (manual / AI-assisted)
- before/after hashes
- proposal record + operator acceptance (AI mode)

Manual edit: Class 1
AI-assisted edit: Class 2

CORE PRINCIPLE: Editable Regions change what the operator can conveniently revise before publication — they change no authority.

Overlay editing is preparation and composition, never execution.